



Read like a radiologist: Efficient vision-language model for 3D medical imaging interpretation

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ABSTRACT

Recent medical vision-language models (VLMs) have shown promise in 2D medical image interpretation. However extending them to 3D medical imaging has been challenging due to computational complexities and data scarcity. Although a few recent VLMs specified for 3D medical imaging have emerged, all are limited to learning volumetric representation of a 3D medical image as a set of sub-volumetric features. Such process introduces overly correlated representations along the z-axis that neglect slice-specific clinical details, particularly for 3D medical images where adjacent slices have low redundancy. To address this limitation, we introduce MS-VLM that mimic radiologists' workflow in 3D medical image interpretation. Specifically, radiologists analyze 3D medical images by examining individual slices sequentially and synthesizing information across slices and views. Likewise, MS-VLM leverages self-supervised 2D transformer encoders to learn a volumetric representation that capture inter-slice dependencies from a sequence of slice-specific features. Unbound by sub-volumetric patchification, MS-VLM is capable of obtaining useful volumetric representations from 3D medical images with any slice length and from multiple images acquired from different planes and phases. We evaluate MS-VLM on publicly available chest CT dataset CT-RATE and in-house rectal MRI dataset. In both scenarios, MS-VLM surpasses existing methods in radiology report generation, producing more coherent and clinically relevant reports. These findings highlight the potential of MS-VLM to advance 3D medical image interpretation and improve the robustness of medical VLMs.

1. Introduction

Attempts to non-invasively visualize the internal structures of the human body have driven significant advancements in medical imaging technology. These efforts began with two-dimensional (2D) modalities like plain X-rays, but gradually evolved toward more sophisticated three-dimensional (3D) imaging techniques, such as computed tomography (CT) and magnetic resonance imaging (MRI), which offer a more detailed visualization of anatomical structures (Bushberg and Boone, 2011; Kalender, 2011; McRobbie et al., 2017). CT and MRI imaging, by providing comprehensive insights without the need for invasive procedures, have become crucial and indeed inevitable tools in modern medicine, supporting both diagnosis and treatment across

diverse medical fields (Bushberg and Boone, 2011; Kalender, 2011; McRobbie et al., 2017).

However, interpreting 3D medical imaging modalities, such as CT and MRI demands high level of expertise, often requiring significant time and effort from well-trained radiologists. Radiologists, however are not immune to the risk of error (Brady, 2017), which has spurred the development of artificial intelligence (AI) models aimed at assisting radiologists' 3D medical image interpretation (Litjens et al., 2017). Despite some successes, particularly in specific tasks such as segmentation and detection (Wang et al., 2019; Hatamizadeh et al., 2022; Litjens et al., 2017), when compared to the performance of multi-modal models on simpler 2D modalities like X-rays (Lee et al., 2023; Li

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et al., 2024; Chaves et al., 2024), these AI models still face significant limitations in broader applications such as radiology report generation or visual question answering (VQA).

The limited success of 3D imaging AI models can be attributed to several key factors. First, there is a fundamental shortage of paired 3D image-report datasets compared to the abundance of datasets available for 2D imaging modalities such as X-rays (Johnson et al., 2019; Demner-Fushman et al., 2012; Irvin et al., 2019), resulting in significant data insufficiency. This challenge is further exacerbated by the immense data volume and computational demands of 3D imaging, which are several hundred times greater than those of 2D imaging (Litjens et al., 2017). Second, the inherent complexity of 3D medical imaging poses significant challenges for effective encoding, as current 3D vision encoders often lack the capacity to capture the full scope of information required for accurate 3D image interpretation (Litjens et al., 2017; Dou et al., 2017; Isensee et al., 2021). Existing vision-language models (VLMs) also struggle to efficiently process the complex and voluminous data inherent in 3D imaging, leading to notable inefficiencies. Specifically, most state-of-the-art models (Wu et al., 2023; Blankemeier et al., 2024; F. Bai et al., 2024) rely on 3D vision encoders that partition image volumes into fixed-size 3D patches. This approach introduces processing inefficiencies by creating unnecessary correlations across the z-axis, further limiting their effectiveness in handling 3D medical imaging data.

To address these challenges, we propose a novel VLM inspired by the way radiologists interpret 3D medical images. Unlike traditional methods that treat the entire 3D volume as a uniform entity, our model, termed MultiSlice Vision Language Model (MS-VLM), processes 3D volumes as a collection of 2D planes. This approach enables the model to encode information from each 2D plane independently before combining them to produce a comprehensive interpretation. By leveraging this method, MS-VLM achieves significantly higher data and computational efficiency while delivering superior performance compared to conventional approaches. Additionally, the structural design of MS-VLM allows it to handle volumetric data with variable slice lengths, enhancing its versatility and adaptability across diverse clinical applications. Notably, MS-VLM excels in scenarios where specific planes and phases are critical for accurate diagnosis, such as rectal MRI, where interpretations across different phases (e.g., pre-contrast and post-contrast) and planes (e.g., sagittal and axial) are essential. By incorporating reconstructions of images from multiple planes and phases, MS-VLM closely mirrors the interpretive processes employed by radiologists, leading to improved diagnostic performance.

Our contributions can be summarized as follows:

- Inspired by radiologists, we propose a novel 3D imaging interpretation method that treats 3D volumes as a collection of 2D planes, significantly improving data and computational efficiency.
- By introducing a novel Z-former, the proposed MS-VLM model handles volumetric data with variable slice lengths, enhancing its practical utility in diverse clinical scenarios.
- Specifically, by incorporating reconstructions from various planes and phases through the Z-former, MS-VLM aligns with radiologists' interpretative workflows, resulting in improved performance.

2. Related work

In this section, we provide a concise overview of recent research on the development of multi-modal foundation models in the medical domain and the approaches of the VLM for 3D medical imaging.

2.1. Multi-modal foundation model in medical domain

Healthcare is inherently multi-modal, requiring clinicians to synthesize information from various sources, such as medical images, clinical notes, lab results, electronic health records (EHR), genomic data, etc. to provide effective patient care. In recent years, significant progress has been made in developing AI systems that achieve expert level performance within individual modalities, such as interpreting radiographic images, analyzing pathology slides, or detecting genetic mutations (Pham et al., 2022; Taori et al., 2023; Qu et al., 2021). However, the complexity and multi-modal nature of healthcare require AI models capable of integrating these diverse data types into a cohesive framework. The early attempts to achieve this integration leverage contrastive learning models, such as CLIP (Radford et al., 2021). CLIP maps images and corresponding contextual information into a shared embedding space through large-scale contrastive learning, demonstrating promising results in integrating simple medical imaging data, like chest radiographs, with radiology reports. Medical adaptations of CLIP such as MedCLIP, CheXzero and CXR-CLIP (Wang et al., 2022; Tiu et al., 2022; You et al., 2023) perform image-text retrieval on chest X-ray (CXR) images and the corresponding reports demonstrating bidirectional understanding of the two modalities.

Building on this foundation, more sophisticated approaches emerged to directly integrate data from different modalities. Unlike earlier CLIP-based methods, these newer approaches enable a more comprehensive understanding of multi-modal data, supporting advanced applications such as detailed medical report generation and VQA. Vision encoders for image data and text encoders for textual information are used together, with cross-attention and self-attention mechanisms facilitating seamless integration of complex information. This deeper level of integration not only correlates visual and textual data but also supports holistic tasks that require a nuanced understanding of medical data. Notable examples include PPKED, MedViLL, and Medical X-VL (Liu et al., 2021; Moon et al., 2022; Park et al., 2024).

More recently, the impressive capabilities of large language models (LLMs) have driven efforts to develop multi-modal models with a generalist architecture. These models aim to handle multiple modalities using a single framework, enhancing flexibility and simplifying the integration process compared to specialized encoder modules. Med-PaLM M (Tu et al., 2024) is a pioneering effort in this area, demonstrating the ability to process text, imaging, and genomic data within a unified framework. This design maximizes flexibility and information transfer across modalities, providing a strong foundation for future multi-modal AI systems. Med-PaLM M has also shown potential to outperform specialized models in specific tasks, suggesting that generalist models may offer superior performance even in domain-specific applications. Subsequent models, such as BioMedGPT and PathChat (K. Zhang et al., 2024; Lu et al., 2024), have further built on these capabilities, underscoring the potential of foundational models to provide generalized solutions in the medical domain.

2.2. VLMs for 3D medical imaging interpretation

Most initial efforts to develop effective multi-modal models in the medical domain have largely focused on simpler 2D imaging modalities, such as plain radiographs and pathology slides (Tu et al., 2024; Li et al., 2024). However, developing such models for 3D medical imaging modalities, like CT and MRI, have presented unique challenges: limited training data and high data dimensionality. Furthermore, current vision encoders often lack the capacity to adequately process the intricate information found in 3D medical images, as they require handling large-scale data while understanding complex spatial relationships.

Unlike natural 3D data such as video, which contains significant redundancy across numerous frames and can comprise thousands of individual frames over a short duration, medical 3D imaging typically consists of fewer slices along the z-axis, usually ranging from 20 to 300.

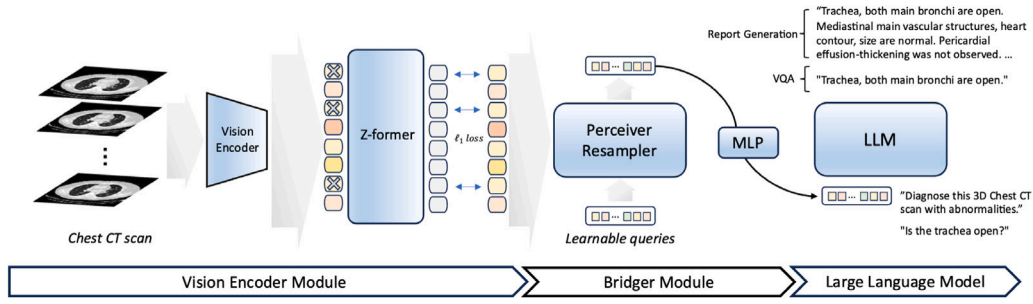


Fig. 1. Overview of MS-VLM architecture.

This fundamental difference presents challenges for the application of fixed-size 3D patch embedding methods. Approaches using fixed-size patches, such as those with dimensions of 16 by 16 by 16, often struggle to capture fine anatomical details effectively due to the lower number of slices in the z-axis compared to the high in-plane resolution typical of medical imaging, such as 512 by 512 for CT scans. As a result, while fixed embedding methods have proven effective for processing natural 3D data, they are less suited to the structural and informational characteristics of medical 3D imaging.

Despite these challenges, progress has been made. RadFM (Wu et al., 2023) supports both 2D and 3D images, focusing on text generation tasks like report generation, rationale generation and VQA. Merlin (Blankemeier et al., 2024) pre-trains a 3D (I3D) ResNet152 to learn volumetric representation applied to downstream tasks such as report generation and segmentation. Another significant contribution is CT-CLIP (Hamamci et al., 2024a), introducing the open-source dataset CT-RATE, which consists of over 25,000 chest CT volume-report pairs. CT-CLIP demonstrates that contrastive learning can align CT images with reports using a 3D vision encoder and text encoder, similar to CLIP. Both CT2Rep (Hamamci et al., 2024b) and 3D-CT-GPT (Chen et al., 2024) employ the 3D vision encoder architecture from CT-CLIP as part of its chest CT report generation model. Unlike CT2Rep that leverages a simple transformer decoder for report generation only, 3D-CT-GPT leverages LLM and perform both report generation and VQA. The M3D-LaMed model (F. Bai et al., 2024) further advances the field by integrating visual information into LLM through a perceiver module, enabling tasks like captioning, VQA, and segmentation. However, M3D-LaMed still relies on a 3D Vision Transformer (ViT) (Dosovitskiy, 2020) with fixed patch embedders, limiting its ability to capture detailed visual information, and retaining challenges in representing complex 3D medical structures effectively.

3. Method

3.1. Model architecture

MS-VLM comprises a vision encoder module, a bridger module, and a LLM as shown in Fig. 1. The vision encoder includes a 2D ViT and a novel Z-former that integrates slice-by-slice 2D ViT embeddings, while the bridger module features a perceiver resampler and an MLP layer to project volumetric representation to a fixed-size learnable query. The learnable query is then used as a visual prompt when instruction fine-tuning the LLM to perform vision-language tasks (report generation or VQA).

Given a 3D medical image, each slice along the z-axis is processed by a pre-trained 2D ViT. The [CLS] token embeddings from the last layer are then concatenated to form a volumetric representation of the given image as a sequence of slice-specific features. This volumetric representation is input to Z-former, which is constructed using a conventional transformer encoder architecture that employs sparse attention mechanism, Big Bird (Zaheer et al., 2020). The last hidden layer of Z-former then outputs a volumetric representation as a sequence of

sub-volumetric features of which each feature incorporate information from both the original and neighboring slice-specific features. The last hidden layer output of Z-former is then used as keys and values for the perceiver resampler. The learned queries from the perceiver resampler then become the final volumetric representation of the given image and is provided as a visual prompt when instruction fine-tuning the LLM for report generation or VQA. The perceiver resampler allows the model to handle 3D medical images of varying slice lengths by enabling LLM fine-tuning on a fixed-size learnable query.

3.2. Model training

The training consists of four stages — Stage 0: Training a domain-specific DINO vision encoder, Stage 1: Training a Z-former, Stage 2: Training a bridger module; Stage 3: Instruction fine-tuning a LLM for report generation and VQA. Fig. 2 demonstrates the overall scheme of how the vision and language data are processed for each stage of training.

3.2.1. Training a domain-specific DINO vision encoder

We utilize a 2D ViT-B/16 model pre-trained on ImageNet-1K using DINO (Self-Distillation with No Labels) (Caron et al., 2021) to extract slice-specific spatial features from volumetric images. DINO is a self-supervised learning method that leverages a teacher-student network structure to learn visual representations without labels. By aligning information between views, DINO effectively captures high-quality features suitable for various downstream tasks. During the self-supervised learning process, DINO enables the 2D ViT to learn semantic representations and essential patch-level features, which are crucial for visual understanding. For a volumetric image with slice length L , each 2D slice i is processed through the pre-trained encoder. We extract the [CLS] token embedding $\mathbf{z}_i^{[\text{CLS}]}$ from the final layer, representing slice-specific spatial features. These embeddings are concatenated along the z-axis to create a comprehensive volumetric representation.

$$\mathbf{Z}_{\text{vol}} = [\mathbf{z}_1^{[\text{CLS}]}, \mathbf{z}_2^{[\text{CLS}]}, \dots, \mathbf{z}_L^{[\text{CLS}]}]$$

The concatenated embeddings $\mathbf{Z}_{\text{vol}}$ are then input to the Z-former for further processing.

3.2.2. Training the Z-former

To train the Z-former, we introduce masked embedding modeling (MEM), inspired by masked image modeling (Xie et al., 2022). Instead of predicting pixel values, MEM regresses on [CLS] token embeddings from $\mathbf{Z}_{\text{vol}}$, where each embedding is masked with a fixed probability. The Z-former reconstructs the masked embeddings, optimizing via ℓ_1 loss against the original:

$$\mathcal{L}_{\text{MEM}} = \frac{1}{|M|} \|\hat{\mathbf{Z}}_{\text{masked}} - \mathbf{Z}_{\text{vol}}\|_1 \quad (1)$$

Here, M refers to the set of masked embeddings and $|M|$ denotes its cardinality. The ℓ_1 norm is defined as the sum of the absolute differences of each vector embedding.

Overall Scheme

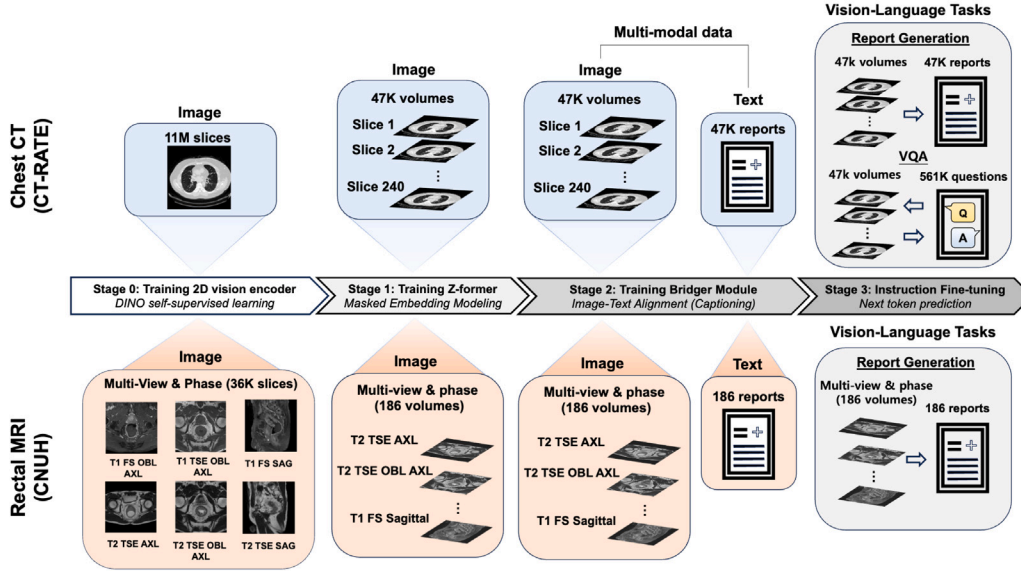


Fig. 2. Overview of MS-VLM training and evaluation. (a) MS-VLM is evaluated on public chest CT dataset CT-RATE and in-house Rectal MRI dataset. MS-VLM learns volumetric representation off of a single axial view for chest CT and multi-view for rectal MRI.

To mimic radiologists' workflow and reduce computational overhead, we adopt the sparse attention mechanism, BigBird for the Z-former. BigBird achieves linear time complexity by combining three attention strategies: sliding window attention, which captures local dependencies among neighboring slices—critical for volumetric imaging as adjacent slices often share anatomical or pathological features; global attention, where designated global tokens attend to all others (and vice versa), ensuring effective modeling of long-range contextual information across the entire volume; and random attention, which enables each token to attend to a small set of randomly selected tokens, thereby preserving the ability to learn long-range interactions without incurring quadratic costs. Other sparse architectures like Longformer (Beltagy et al., 2020) and Performer (Choromanski et al., 2020) also achieve $O(n)$ complexity but lack this full complement of attentional mechanisms: Longformer omits random attention, reducing its capacity for global dependency capture, while Performer and Linformer (Wang et al., 2020) forego sliding-window attention, diminishing their ability to model local, slice-wise continuity. Consequently, by using BigBird, the Z-former effectively captures both local and global spatial relationships along the z -axis within each volume. The final hidden state thus encodes rich sub-volumetric features informed by the current slice, its neighbors, and distant regions.

3.2.3. Training the bridger module

The task-agnostic volumetric representation generated by the Z-former requires further alignment with the language space of the LLM. To address this modality gap, we utilize a perceiver resampler (Alayrac et al., 2022) and an MLP with ReLU activation. The Z-former's final hidden state is projected as keys and values for the perceiver resampler's cross-attention module, where the learnable query attends to specific sub-volumetric representations and aggregates them into refined visual features. These features are subsequently mapped to the LLM's conceptual language space using the MLP, which adjusts the query dimension to match that of the LLM embedding space.

During this process, the DINO vision encoder, Z-former and LLM remain frozen, ensuring the preservation of pre-trained knowledge in the vision encoder module while avoiding catastrophic forgetting in the LLM. The bridger module is optimized to align the visual feature to the LLM embedding space such that aligned feature generate the corresponding reports. The bridger module thus plays a critical role in transforming the visual features into a format that is fully compatible with the LLM's language space.

3.2.4. Instruction fine-tuning

We employ Vicuna-7B-v1.5 (Chiang et al., 2023) as our LLM, selected for its exemplary instruction-following capabilities among publicly accessible checkpoints (Liu et al., 2024; Chiang et al., 2023; Taori et al., 2023). The instructions provided for the chest CT report are illustrated in Fig. 3.

The placeholder <ImageHere> represents the learned query generated by the bridger module, which will henceforth be referred to as the visual features. These visual features are encapsulated within and token embeddings to distinguish them from the report generation instructions. During this stage, the DINO vision encoder and the Z-former remain frozen, while only the bridger module and the LLM are subject to fine-tuning. Specifically, the LLM is fine-tuned using Low-Rank Adaptation (LoRA) by minimizing the negative log-likelihood (NLL) of generating the report $Y = \{y_1, y_2, \dots, y_T\}$, conditioned on the visual features Q and the instruction I .

$$\mathcal{L}_{\text{inst}} = - \sum_{t=1}^T \log P(y_t | y_{<t}, Q, I) \quad (2)$$

One key challenge in aligning 3D medical images, such as CT scans, with radiology reports is the presence of multiple findings within a single scan. Radiology reports inherently describe diverse findings, complicating effective alignment between the image and the text, often resulting in inadequate representation of individual findings.

To address this challenge, we decompose radiology reports into individual components and construct a VQA dataset. Each VQA pair specifically queries the presence and attributes (e.g., location or severity) of a single finding, thereby ensuring precise alignment between each distinct finding in the image and its corresponding textual description. For this purpose, we utilize the RadGenome-ChestCT dataset (X. Zhang et al., 2024), which provides VQA pairs related to the presence, location, size, and type of abnormalities in CT-RATE chest CT scans. Additionally, we generate synthetic VQA pairs based on CT-RATE reports using LLaMA-3-8B, as illustrated in Fig. 4, with detailed prompt instructions available in Appendix A. A combination of RadGenome-ChestCT train dataset and the synthetic VQA pairs are used for training MS-VLM for the VQA task.

Finally, MS-VLM is fine-tuned on both the report generation task and the VQA task. The report generation task aims to generate comprehensive radiology reports, while the VQA task focuses on answering specific questions about individual findings. These tasks are jointly fine-tuned with a 1:1 sample ratio per training iteration.

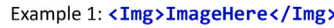
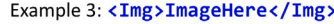
Example 1:  What abnormalities do you see from this 3D chest CT scan?
 Example 2:  Please provide a detailed description of this 3D chest CT scan.
 Example 3:  Diagnose this 3D chest CT scan with any abnormalities.

Fig. 3. Instruction prompts used for instruction fine-tuning the LLM for report generation.

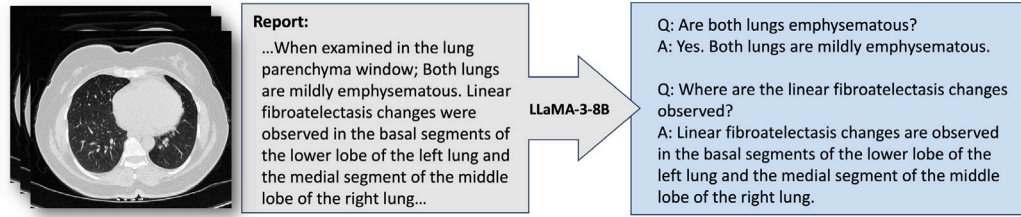


Fig. 4. Generating synthetic VQA pairs from CT-RATE reports using LLaMA-3-8B.

3.3. Implementation details

3.3.1. Datasets

For the chest CT report generation task, we evaluate our model using the CT-RATE dataset, introduced in CT-CLIP (Hamamci et al., 2024a). The CT-RATE dataset comprises 25,692 non-contrast chest CT volumes from 21,304 unique patients. With multiple reconstructions, the dataset expands to 50,188 volumes, each paired with radiology reports detailing findings and impressions.

We train our model on 47,149 volumes from 20,000 unique patients, excluding cases with insufficiently short reports. The test set consists of 3039 volumes from 1304 unique patients. To ensure a fair comparison with existing methods, such as CT2Rep, we focus exclusively on the findings sections of the reports.

CT volumes are preprocessed following the CT-CLIP protocol (Hamamci et al., 2024a), which includes resizing to achieve uniform spacing of 1.5 mm along the z-axis and 0.75 mm along the x- and y-axes. Each volume is then padded or center-cropped to maintain a consistent resolution of $480 \times 480 \times 240$ across the dataset.

For the chest CT VQA task, we utilize both our synthetic VQA dataset and the RadGenome-ChestCT train dataset (X. Zhang et al., 2024) for training and only the RadGenome-ChestCT validation dataset for evaluation. The synthetic dataset, comprising approximately 180,000 QA pairs, excludes instances with overly brief answers, such as “Yes” or “No”. RadGenome-ChestCT, a large-scale dataset with detailed 3D chest CT annotations, provides 1.3 million QA pairs based on CT-RATE. For training, from the RadGenome-ChestCT train dataset, we filter QA pairs to retain only those mentioning the most specific anatomical regions associated with common abnormalities, ensuring precise alignment. An example QA pair is provided in Appendix A. From the filtered QA pairs, we randomly select 121,000 QA pairs related to the types of abnormalities, 136,000 presence-related QA pairs, 110,000 location-related QA pairs, and 14,000 size-related QA pairs, creating a comprehensive training dataset that ensures robust coverage across different aspects of chest CT interpretation. For evaluation, we likewise filter QA pairs from the RadGenome-ChestCT validation dataset, to retain only those mentioning the most specific anatomical regions associated with common abnormalities. We then use all the filtered QA pairs to evaluate the entire CT-RATE validation dataset.

The rectal MRI report generation was conducted using in-house data obtained from Chungnam University Hospital. The dataset comprised MRI volumes from 311 patients and their corresponding radiology reports. Most of the radiology reports, though unstructured and presented in free-form text, consistently contained information on seven key findings: tumor location, peritoneal involvement, T staging, circumferential resection margin (CRM) involvement, anal sphincter involvement (ASI), mesorectal lymph node involvement (MLNI), and extramural venous invasion (EMVI). Tumor location was measured in centimeters from the

anal verge to guide treatment decisions, while peritoneal involvement was categorized as none, partial, or full to assess the extent of invasion. T staging classified the tumor’s depth (T1–T4), indicating its local invasiveness. The CRM was evaluated to determine the proximity of the tumor to the mesorectal fascia, which serves as an indicator of local recurrence risk. ASI assessed the potential tumor extension into the anal sphincter, influencing surgical options regarding sphincter preservation. MLNI involved checking for signs of lymph node enlargement or metastasis, which are crucial prognostic factors. EMVI assessed the tumor’s invasion into extramural veins, which is associated with an increased risk of metastasis (refer to Fig. 5 for details).

The MRI volumes for each patient were acquired using three different contrasts: T1-weighted (T1), T2-weighted (T2), and Diffusion-Weighted Imaging (DWI). The T1 and T2 volumes were obtained across five imaging planes: axial, oblique axial, sagittal, coronal, and oblique coronal, while the DWI volumes were captured exclusively in the axial plane. The T1 volumes were acquired with or without modifiers such as Fat Suppression (FS), Spectral Saturation with Inversion Recovery (SPAIR), and Contrast Enhancement (CE) to improve image clarity. For some patients, both modified and unmodified versions of T1 volumes were available. Overall, the MRI volumes per patient included at least four axial volumes, two sagittal volumes, and one coronal volume.

These multi-view and multi-phase MRI volumes were utilized to compare a baseline model that exclusively uses T2 axial volumes, which are considered most critical for rectal tumor MRI interpretation, with the proposed MS-VLM, capable of robustly processing all imaging phases regardless of sequence length. Further details regarding the availability of volumes for each imaging plane and contrast phase can be found in Appendix B.

The study received approval from the Institutional Review Board (IRB) of Chungnam University Hospital (IRB number 2022-05-113), and the requirement for informed consent was waived due to the retrospective nature of the study.

3.3.2. Training details

To train on CT-RATE during Stage 0, we employed a ViT-B/16 model as the DINO vision encoder, initialized with ImageNet-1K DINO pretrained weights. We fine-tuned the model for 50 epochs on volumetric image slices using DINO. Global and local crops were applied with area ranges of 0.8–1.0 and 0.1–0.3, respectively, relative to the input image area. Data augmentations included random rotations, auto-contrast, and equalization.

For subsequent training stages, input image dimensions were set to $240 \times 3 \times 480 \times 480$, representing full volumes. A Z-former, with 12 hidden layers and a model dimension of 768, was employed. The Z-former utilized a sparse attention mechanism with a sliding window size of 16 and three random blocks, and two global blocks. It was pre-trained for 20 epochs using masked embedding modeling (MEM) to capture

"The distance of the lowest tumor margin from the anal verge measures about 3 cm. There is no relationship between the tumor and the peritoneum. The tumor is partially encircling in its circumferential location and measures about 5 cm longitudinally. T-staging is CT3, indicating a thickness greater than 5 mm. The circumferential resection margin involvement is positive, and the anal sphincter is absent. Mesorectal lymph nodes are absent, as are extramural lymph nodes. Extramural venous invasion (EMVI) is also absent."

Tumor Location; Peritoneal Involvement; T-staging; Circumferential resection margin (CRM) involvement; Anal Sphincter Involvement(ASI); Mesorectal lymph node involvement (MLNI); Extramural venous invasion (EMVI)

Fig. 5. An example of a rectal MRI report that describes the seven most common rectal tumor-related abnormalities. Each sentence is color-coded based on the abnormality it describes.

dependencies along the z-axis. The masking probability was set to 0.3. Both the 2D ViT and Z-former were trained with the Adam optimizer with learning rate of $1e-4$ and an initial warmup phase of 50 linear iterations starting from $1e-5$.

The perceiver resampler was initialized with 32 learnable queries, each with a dimension of 4096. Both the perceiver resampler and an associated MLP layer were randomly initialized before training. For the LLM, Vicuna-7B-1.5v was used. The bridger module was trained for 1 epoch, followed by 5 epochs of joint training with the LLM. An AdamW optimizer was used with a learning rate of $1e-4$ and a weight decay of 0.05.

The rectal MRI dataset consisted of 311 patients, of which 82 were excluded from training due to radiology reports lacking sufficient detail on the seven key findings. The remaining dataset was divided into 186 patients for training and 43 patients for testing. For all training stages, we employed a configuration similar to the CT-RATE training pipeline. Specifically, a domain-specific DINO vision encoder was pre-trained in a self-supervised manner using only the T1/T2 axial and sagittal views, as these are the most critical planes for analyzing rectal tumors. In cases where patients had more volumes than the required number, a prioritization strategy was applied to select volumes in the following order: T2 volumes, T1 volumes with modifiers (e.g., FS, CE), and T1 volumes without modifiers. This order reflects the relative significance of these imaging modalities in diagnosing rectal tumors. As a result, 20 slices were sampled from each of the six selected phases, leading to a stacked input configuration of $120 \times 3 \times 480 \times 480$.

All experiments on CT-RATE were conducted on eight NVIDIA A100 (40 GB) GPUs, while those on rectal MRI were conducted on a single NVIDIA A100 (40 GB) GPU.

3.4. Evaluation metrics for chest CT

To assess the clinical efficacy of the generated radiology reports, we employ two evaluation approaches: natural language generation (NLG) metrics, including BLEU (Papineni et al., 2002), ROUGE (Rouge, 2004), and METEOR (Lavie and Denkowski, 2009), and clinical accuracy (CA) metrics, including precision, recall and F1, as utilized in CT2Rep. NLG metrics quantify the linguistic similarity between the generated reports and the ground truth by evaluating n-gram overlaps, sequence alignment, and synonym usage. These metrics provide an objective measure of the overall linguistic quality and coherence of the generated reports with respect to the ground truth.

For CA metrics, we use a RadBERT-based text classifier introduced in CT-CLIP (Yan et al., 2022; Hamamci et al., 2024a) to determine whether the generated reports include the abnormalities present in the ground truth. We calculate precision, recall, and F1 scores for each abnormality to evaluate this alignment. CA metrics thus measure how well the generated reports capture critical findings from the ground truth while minimizing hallucinated content.

Radiology reports may differ in specific phrasing or structure while conveying fundamentally the same information. To accommodate such variability, we employed an additional evaluation framework utilizing LLMs. Conventional NLP metrics, which primarily capture structural

similarity, often fail to account for semantic equivalence. Similarly, CA metrics provide quantifiable measures but struggle to reflect nuanced or alternative expressions of equivalent clinical content. LLM-based evaluation offers a potential solution by addressing these limitations and enabling more flexible and context-aware assessments.

In the evaluation of chest CT reports, four key categories were defined: Presence, Severity, Location, and Hallucination. Each category is assessed binarily, with a score of 1 indicating complete accuracy across all abnormal findings in the reference report. A score of 0 signifies the presence of factual errors in the generated report concerning the relevant category. GPT-4o-mini (Achiam et al., 2023) was used for evaluation. Detailed evaluation prompts for each category are provided in Appendix C.

3.5. Evaluation metrics for rectal MRI

The evaluation of rectal MRI involved the seven key findings previously mentioned to assess tumor staging and invasion: tumor location, peritoneal involvement, T staging, CRM involvement, ASI, MLNI, and EMVI. Given the heterogeneous nature of these findings, distinct evaluation approaches were employed based on their characteristics.

Metrics such as tumor location and peritoneal involvement are inherently complex and difficult to evaluate using straightforward binary classification. These findings often involve nuanced descriptions and categorizations, such as precise tumor distance from the anal verge or the extent of peritoneal invasion. To address this complexity, a LLM was utilized to indirectly assess the validity of these metrics. By leveraging advanced contextual understanding, the LLM provided an approximate measure of accuracy. Specifically, o1-mini (Achiam et al., 2023) was employed for this evaluation, enabling a flexible and context-aware assessment of these findings. The prompts used for o1-mini evaluation is detailed in Appendix C.

In contrast, findings such as CRM involvement, ASI, MLNI, and EMVI, which can be distinctly classified as either positive or negative, were evaluated using CA metrics. These metrics quantified the classification accuracy through measures such as precision, recall, and F1 scores, providing a robust evaluation of the generated reports' ability to accurately reflect the presence or absence of these clinically significant features.

4. Experimental results

4.1. Experimental results on CT-RATE

Among the existing methods for chest CT report generation, we selected CT2Rep and 3D-CT-GPT as primary baselines due to their demonstrated effectiveness in processing chest CT scans. Other approaches, such as RadFM and M3D-LaMed, rely on pre-training datasets sourced from Radiopedia, and their performance on report generation has been shown to lag behind that of 3D-CT-GPT. Accordingly, our evaluation emphasizes models that are specialized for chest CT report generation.

Using the official implementation and training settings provided for CT2Rep, we successfully reproduced its results and evaluated its

Table 1

Comparison of NLG metrics and CA metrics on the generated reports between MS-VLM and baseline methods.

Method	NLG metrics			CA metrics		
	BLEU-4	ROUGE-L	METEOR	Precision	Recall	F1 score
3D-CT-GPT	0.133	0.125	0.148	<u>0.239</u>	0.195	0.190
Merlin	0.279	0.431	0.437	0.202	0.223	0.214
CT2Rep (3D encoder)	<u>0.252</u>	<u>0.432</u>	0.328	0.355	0.132	0.175
MS-VLM (3D encoder)	0.184	0.364	0.331	0.193	0.231	0.207
MS-VLM (w/o VQA)	0.189	0.374	0.313	0.192	<u>0.298</u>	<u>0.232</u>
MS-VLM (proposed)	0.232	0.438	<u>0.396</u>	0.222	0.329	0.261

The best performance is highlighted in bold, and the second-best is underlined. CA metrics are averaged across 18 chest-related abnormalities.

performance on the CT-RATE test set. Since the code for 3D-CT-GPT is not publicly available, we re-implemented 3D-CT-GPT by combining the pre-trained CT-ViT (Hamamci et al., 2024a) as the vision encoder, a linear layer as a projection layer and Vicuna-7B-v1.5 as the LLM.

In addition, to demonstrate the efficiency of MS-VLM's volumetric representation processing, we conducted an additional experiment where the vision encoder module in MS-VLM was replaced with a 3D ViT, closely mirroring the architecture used in existing models. We refer to this MS-VLM variant as MS-VLM (3D encoder). To train MS-VLM (3D encoder), we first freeze the LLM and optimize the 3D ViT and the bridger module for image-to-text alignment. Then the entire model is fine-tuned end-to-end for report generation. A direct comparison with MS-VLM (3D encoder) under identical experimental conditions enables a clear evaluation of the improvements introduced by our approach.

We also added Merlin as a baseline to compare MS-VLM to recent medical vision-language models that utilize pre-trained 3D vision encoder. To measure the performance of chest CT report generation of Merlin, we pre-trained Merlin with a combination of InfoNCE loss and binary cross entropy loss. For training with InfoNCE loss, we used pairs of CT-RATE volumes and their corresponding reports while for training with binary cross entropy loss, we used the CT-RATE volumes and their multi-abnormality class labels. Once trained, the last hidden layer embeddings of Merlin were projected to Vicuna-7B-v1.5 using a trainable linear layer for image-to-report alignment. We then further fine-tuned the linear layer and Vicuna-7B-v1.5 for report generation.

4.1.1. Report generation results

As presented in Table 1, MS-VLM achieves the highest overall performance in terms of CA metrics, with an average F1 score of 0.261. While CT2Rep demonstrates higher precision, its lower recall reduces its overall F1 score. The qualitative comparison in Fig. 6 further highlights the superiority of MS-VLM. It effectively recalls a greater number of abnormal findings, generating reports that closely align with the ground truth. Notably, MS-VLM demonstrates the ability to identify subtle positive findings, such as satellite masses and fibrotic changes, which are often overlooked by other baseline models.

MS-VLM also outperforms its variant, MS-VLM (3D encoder), which is a variant of MS-VLM using 3D ViT encoder without Z-former, in both NLG and CA metrics. Higher average ROUGE, METEOR, and F1 scores confirm that MS-VLM generates reports that better match radiologists' annotations while addressing critical findings more effectively, demonstrating the utility of the proposed combination of a 2D vision encoder and the Z-former compared to conventional 3D vision encoder approaches. As shown in Fig. 6, MS-VLM identifies more true positives compared to MS-VLM (3D encoder). For additional qualitative analyses, refer to Appendix D. Similar trends are observed in comparisons with other MS-VLM variants. When VQA task is not performed jointly with report generation task during training, MS-VLM (w/o VQA) performance becomes suboptimal, suggesting the importance of integrating both tasks for optimal performance. Merlin excels in terms of NLG metrics, however most of the generated reports overfit to the reports that only carry normal findings, which are most frequent in the training data. Therefore, despite superior NLG metric performance, Merlin fails to achieve as high CA metric performance as our proposed MS-VLM.

Table 2

Comparison of GPT-4o-mini evaluation performance between MS-VLM and baseline methods.

Method	Presence	Location	Severity	Hallucination
3D-CT-GPT	0.182	0.193	0.216	0.124
CT2Rep (3D encoder)	0.186	0.197	0.216	0.186
MS-VLM (3D encoder)	0.203	<u>0.216</u>	<u>0.223</u>	0.122
MS-VLM (w/o VQA)	<u>0.212</u>	0.207	0.209	0.124
MS-VLM (proposed)	0.217	0.225	0.234	<u>0.127</u>

The best performance is highlighted in bold, and the second-best is underlined.

Table 3

Comparison of CA metrics of the generated reports when MS-VLM is given volumes with fixed z-length versus volumes with original slice length.

Method	Precision	Recall	F1
All volumes			
3D-CT-GPT ($z = 240$)	<u>0.239</u>	0.195	0.190
CT2Rep ($z = 240$)	0.355	0.132	0.175
MS-VLM ($z = 240$)	0.222	0.329	0.261
MS-VLM ($z = L$)	0.234	<u>0.286</u>	<u>0.245</u>
Volumes with $L > 240$			
3D-CT-GPT ($z = 240$)	<u>0.270</u>	0.195	0.207
CT2Rep ($z = 240$)	0.330	0.146	0.190
MS-VLM ($z = 240$)	0.229	<u>0.258</u>	<u>0.237</u>
MS-VLM ($z = L$)	0.226	0.284	0.246

z denotes the number of slices in z-axis, and L denotes original slice length of the volume.

For quantitative evaluation, detailed CA metric results for MS-VLM across 18 abnormality categories are provided in Appendix E. Table 2 presents the evaluation results using GPT-4o-mini. Unlike abnormality-specific evaluations, GPT-4o-mini assesses the overall report's accuracy for each category on a binary scale. Due to this strict evaluation criterion, even minor inaccuracies result in a score of zero, leading to relatively low average scores across all categories for all models.

MS-VLM demonstrates superior performance in the presence, location, and severity categories, with the exception of the hallucination category. These results indicate that MS-VLM not only accurately predicts abnormalities but also exceeds baseline models in identifying their precise location and severity. However, in the hallucination category, MS-VLM achieves slightly lower scores compared to CT2Rep. The relatively better performance of CT2Rep in the hallucination category is likely attributable to its tokenizer, which has been specifically trained on CT-RATE radiology reports.

MS-VLM outperforms both its variants, MS-VLM (3D encoder) and MS-VLM (w/o VQA), across all evaluated categories. These results suggest the superiority of the proposed Z-former-based approach over traditional 3D vision encoder methods. Furthermore, they highlight the critical role of VQA training focused on single abnormalities in improving the model's understanding and alignments for the fine-grained findings, thereby enabling more precise attention to detailed abnormalities and enhancing overall model performance.

4.1.2. Robustness of Z-former

Our model inherently integrates visual information into a single [CLS] token embedding $z_i^{[CLS]}$ irrespective of the z-length, enabling



True Positive / True Negative / False Positive / False Negative

Ground Truth: Trachea, both main bronchi are open. No occlusive pathology was detected in the lumen. Calcified atheroma plaques were observed in the mediastinal main vascular structures. Metallic densities were observed in the sternum and secondary to surgery on the heart. Heart contour, size is normal. Thoracic aorta diameter is normal. Pericardial effusion-thickening was not observed. Thoracic esophageal calibration was normal and no significant tumoral wall thickening was detected. Lymph nodes with a short diameter of up to 9 mm were observed in the mediastinal prevascular area, aortopulmonary window, paratracheal area, and bilateral hilar region. There was no lymph node that reached pathological size in the bilateral axillary region and supraclavicular region. When examined in the lung parenchyma window; In both lungs, increased aeration consistent with centriacinar emphysema was observed. A mass of approximately 62x32 mm in size with irregular spiculated contours was observed in the superior segment of the left lung lower lobe. Satellite masses, the largest of which reached approximately 8 mm in diameter, were observed in the vicinity of the mass. Mild bronchiectatic changes, peribronchial thickenings and intense ground-glass appearances in the lower lobe of the left lung attract attention, especially in the lower lobe of the left lung. A calcified parenchymal nodule with a diameter of approximately 4 mm was observed in the lateral segment of the lower lobe of the right lung. Fibroatelectatic changes were observed in the bases of both lungs. Pleural effusion-thickening was not detected...

CT2Rep: trachea and both main bronchi are normal . no occlusive pathology was detected in the trachea and both main bronchi. peripheral and centrally located ground glass areas are observed in the upper and lower lobes of both lungs and the middle lobe of the right lung and the lingular segment of the upper lobe of the left lung . the described manifestations were evaluated primarily in favor of viral pneumonia. these findings are frequently observed in covid-19 pneumonia . no mass was detected in both lungs ; mediastinal structures cannot be evaluated optimally because contrast material is not given . as far as can be observed heart contour and size are normal . no pleural or pericardial effusion was detected . atheroma plaques are observed in the aorta and coronary arteries . the widths of the mediastinal main vascular structures are normal . there are short lymph nodes less than 1 cm in diameter in the mediastinum and hilar regions . there is no pathological wall thickness increase in the esophagus within the sections . there is a sliding type hiatal hernia at the lower end of the esophagus.

MS-VLM (3D encoder): Trachea, both main bronchi are open. Mediastinal main vascular structures, heart contour, size are normal. Pericardial effusion-thickening was not observed. Thoracic esophageal calibration was normal and no significant tumoral wall thickening was detected. Small lymph nodes with a short axis measuring up to 12 mm are observed in the mediastinum and hilar regions, more prominently on the right. When examined in the lung parenchyma window; Thickness increases up to 65 mm are observed at all levels in both hemithorax, especially pleural leaves and interlobular septa. Nodules of ground glass density are present. A newly developed nodule with lobulated margins of approximately 8x4 mm is observed in series 2 image 93 in segment II in the superior lower lobe of the left lung upper lobe.

MS-VLM: Trachea, both main bronchi are open. Mediastinal major vascular structures, heart contour, size are normal. Thoracic aorta diameter is normal. Pericardial effusion-thickening was not observed. There were lymph nodes in the mediastinum with short axes measuring up to 9 mm and partially calcified. Calcific atheroma plaques were also selected on LAD. On the right, a 5.9 mm diameter nonspecific parenchymal mass adjacent to the middle lobe. A few millimetric nonspecific pulmonary nodules and slight sequela changes are seen in their vicinity in the lingula inferiorly, in the superior segments of both upper lobes of the bilateral lungs. Pleuroparenchymal fibrotic recessions accompanied by interstitium in both diaphragmatic corners are observed.

Fig. 6. Qualitative evaluation on CT-RATE. Each sentence from the report is color coded to identify True Positive, True Negative, False Positive and False Negatives on the findings mentioned in the ground truth report. MS-VLM achieves greater recall on the true positives and is less prone to generating false negatives. MS-VLM (3D encoder): A variant of MS-VLM using 3D ViT encoder without Z-former.

it to process visual information seamlessly across varying z-lengths. This capability offers a significant advantage in clinical settings where slice lengths are not fixed, as scan ranges are determined post-scout view (tomogram) acquisition, resulting in variable slice lengths for each patient. This flexibility contrasts with traditional models that rely on 3D vision encoder with a fixed z-length, which may lead to information loss.

To validate this hypothesis, we conducted experiments analyzing the impact of slice count on CT volume performance. For a fair comparison with previous studies, the baseline setting involves fixing the z-length to a constant value of 240. However, when the z-length L exceeds 240, fixed z-length approaches may suffer from significant information loss. We compared our method, MS-VLM which uses original slice length without fixing the z-length, against both traditional fixed z-length methods and MS-VLM with fixed z-length.

Table 3 summarizes the experimental results. The MS-VLM approach, which retains the original slice length without imposing a fixed z-length, exhibits no significant performance degradation, demonstrating its adaptability. This advantage is particularly evident in subgroup analyses of data with longer slice length ($L > 240$), where the proposed method outperforms not only fixed z-length approaches such as CT2Rep and 3D-CT-GPT but also the fixed z-length variant of MS-VLM. These findings effectively validate our hypothesis and suggest the utility of MS-VLM in clinical scenarios involving variable z-lengths.

We conducted ablation studies on the Z-former module to evaluate the significance of Masked Embedding Modeling (MEM) for volumetric representation learning. As shown in Table 4, we compare the performance of the full MS-VLM model with two ablated variants: MS-VLM (w/o Z-former) and MS-VLM (w/o Z-former + Adap. pool). In MS-VLM (w/o Z-former), the Z-former module is entirely removed. Instead,

Table 4

Comparison of NLG metrics and CA metrics on the generated reports between MS-VLM and baseline methods.

Method	NLG metrics			CA metrics		
	BLEU-4	ROUGE-L	METEOR	Precision	Recall	F1 score
MS-VLM (w/o Z-former)	0.186	0.302	0.366	0.189	0.224	0.194
MS-VLM (w/o Z-former + Adap.Pool)	<u>0.199</u>	0.321	<u>0.385</u>	0.173	0.158	0.139
MS-VLM (w/o VQA)	0.189	<u>0.374</u>	0.313	<u>0.192</u>	<u>0.298</u>	<u>0.232</u>
MS-VLM (proposed)	0.232	0.438	0.396	0.222	0.329	0.261

The best performance is highlighted in bold, and the second-best is underlined. CA metrics are averaged across 18 chest-related abnormalities.

the [CLS] tokens extracted from the vision encoder are stacked and directly passed through a linear projection layer before being fed into the bridger module. In MS-VLM (w/o Z-former + Adap. pool), we further modify the above variant by applying adaptive pooling to the final hidden layer token representations of the vision encoder. This results in four pooled token embeddings per slice, which are then used as the slice-level representation. The results indicate that the full MS-VLM model outperforms both ablated variants. Notably, increasing the number of tokens used to represent each slice, as done in the adaptive pooling variant, led to degraded performance on the Clinical Accuracy (CA) metric. This suggests that introducing additional vision tokens may interfere with the model's learnable queries, making it harder to identify which slice representations are most relevant during report generation.

4.1.3. Transferability of MS-VLM

We conducted additional experiments to assess the transferability of the self-supervised MS-VLM vision encoder to chest CT segmentation tasks. Specifically, we integrated the MS-VLM vision encoder into the state-of-the-art volumetric segmentation framework SegVol (Du et al., 2024), replacing its original image encoder. This modified version is referred to as SegVol (MS-VLM). We fine-tuned SegVol (MS-VLM) to generate segmentation masks for the five thoracic structures: heart, left lung, right lung, trachea and pulmonary artery. Ground truth annotations were obtained from the RadGenome-ChestCT dataset, and fine-tuning was performed for 5 epochs using the CT-RATE training dataset. For comparison, we followed the original SegVol training procedure as a baseline, referred to as SegVol (3D MIM). This baseline uses a 3D Vision Transformer pre-trained with SimMIM (Xie et al., 2022). To establish this baseline, we first pre-trained the 3D Vision Transformer on CT-RATE using SimMIM for 50 epochs, and then fine-tuned the entire SegVol model for 5 epochs under the same settings as SegVol (MS-VLM). Additionally, we introduced another variant, SegVol (MS-VLM Scratch), in which the MS-VLM vision encoder is used without pretraining (i.e., randomly initialized). This variant allows us to isolate and evaluate the benefit of the self-supervised pretraining in MS-VLM. As shown in Table 5, SegVol (MS-VLM) outperforms SegVol (MS-VLM Scratch) on all five classes, indicating that the self-supervised pretraining of the MS-VLM vision encoder provides a robust initialization for learning anatomical structure representations. SegVol (3D MIM) achieves the highest average Dice score across the three classes. Compared to SegVol (3D MIM), SegVol (MS-VLM) yields lower Dice scores for the heart, right lung, and trachea, but achieves higher Dice scores for the left lung and pulmonary artery. Such results suggest that, although performance is not uniformly superior across all classes, SegVol (MS-VLM) attains segmentation performance broadly comparable to SegVol (3D MIM) under matched pretraining conditions. This indicates that the MS-VLM vision encoder provides a reasonable prior for vision-centric tasks such as segmentation, and that the MS-VLM vision encoder can serve as a valid backbone for downstream tasks beyond vision-language modeling.

4.1.4. Interpretability of MS-VLM

To evaluate the lesion-awareness of MS-VLM for abnormality prediction, we analyze the attention distribution of the learnable queries

Table 5

Comparison of average DICE score for the five thoracic structures: heart, left lung, right lung, trachea and pulmonary artery.

Method	Heart	Left lung	Right lung	Trachea	Pulmonary artery
SegVol (MS-VLM Scratch)	0.265	<u>0.493</u>	0.372	0.093	<u>0.091</u>
SegVol (3D MIM)	0.362	0.456	0.683	0.123	0.088
SegVol (MS-VLM)	<u>0.267</u>	0.517	<u>0.396</u>	<u>0.103</u>	0.142

The best performance is highlighted in bold, and the second-best is underlined.

Table 6

Average Precision@k and Recall@k for top-k slices (k=40) attended by MS-VLM when correctly predicting the abnormalities present in RexGroundingCT dataset.

Abnormality	Precision@40	Recall@40
Emphysema	0.156	0.560
Lung nodule	0.190	0.491
Peribronchial thickening	0.193	0.220
Interlobular septal thickening	0.125	0.278

over the slice representations at the last layer of the perceiver resampler. Specifically, we examine whether the queries place higher attention on lesion-containing slices when generating accurate abnormality predictions in the report. Visualization examples of the attention distribution across slice representations are provided in Appendix G. Since the CT-RATE dataset does not include slice-level lesion annotations, it is not feasible to compute slice-wise attention localization metrics over the full CT-RATE test set. To address this limitation, we conduct additional experiments on the RexGroundingCT dataset (Baharoon et al., 2025), which is a subset of CT-RATE where radiologists have provided slice-level lesion annotations, resulting in a total of 3142 CT scans.

To avoid information leakage, we retrain MS-VLM on the CT-RATE training split after excluding all scans that appear in RexGroundingCT. We then evaluate this model on RexGroundingCT and select only the cases in which the model correctly predicts the presence of a given abnormality. For each such correctly predicted abnormality, we rank slices by the average attention weight that the learnable queries in the last layer of the perceiver resampler assign to each slice representation. Using the per-lesion segmentation masks provided in RexGroundingCT, we derive binary labels indicating whether each slice contains that abnormality. Based on this ranking, we compute precision@k and recall@k over the top-k attended slices (with $k = 40$ out of a total of 240 slices per volume) with respect to the lesion-containing slices, thereby quantifying how strongly the model's attention is concentrated on lesion-bearing slices.

As summarized in Table 6, MS-VLM exhibits meaningful lesion-aware attention patterns across both focal and non-focal abnormalities. For emphysema, the model achieves a relatively high recall@40 (0.560). This aligns with its clinical nature: emphysema typically manifests as a diffuse parenchymal process that extends over broad regions across many slices, making it easier for the model to achieve high slice-level coverage. In contrast, lung nodules are small, focal lesions that often occupy only a few slices within the entire volume. Accordingly, the recall@40 for lung nodules (0.491) is relatively lower than that of

Table 7

Comparison of NLG metrics and CA metrics on the generated reports between MS-VLM and MS-VLM (EP). MS-VLM (EP) undergoes exemplar prompting with the corresponding segmentation mask as the task-specific feature.

Method	NLG metrics			CA metrics		
	BLEU-4	ROUGE-L	METEOR	Precision	Recall	F1 score
MS-VLM	0.262	0.443	0.437	0.178	0.190	0.179
MS-VLM (EP)	0.346	0.508	0.491	0.238	0.228	0.212

The best performance is highlighted in bold. CA metrics are averaged across 18 chest-related abnormalities.

emphysema, which is reasonable given that the model must identify a small number of positive slices within a large 3D context; achieving high slice-level precision is inherently more challenging for such focal findings.

For peribronchial thickening and interlobular septal thickening, both precision@40 and recall@40 are lower than for emphysema and lung nodules. This behavior is also clinically plausible. These findings usually appear as subtle thickening patterns that follow continuous anatomical structures, such as the bronchi or interlobular septa, across consecutive slices. Even for expert radiologists, it is often ambiguous to define the exact starting and ending slices where these abnormalities are present, and substantial interobserver variability are reported for such findings. In this context, it is likely that MS-VLM shows reduced slice-level localization metrics for these subtle, boundary-ambiguous abnormalities. Nevertheless, the non-negligible precision@40 and recall@40 values for these categories indicate that the model's attention is not random and that it retains a measurable degree of sensitivity to these lesion patterns, despite their inherent difficulty.

4.1.5. Segmentation-guided report generation

We hypothesize that segmentation guidance can improve report accuracy as segmentation features provide additional lesion-aware visual cues that help refine the alignment between the input volume and the generated report. To verify this hypothesis, we train a segmentation-guided variant of our model, which applies Exemplar Prompting (Y. Bai et al., 2024) using RexGroundingCT segmentation masks as task-specific visual features on top of the RexGroundingCT-fine-tuned MS-VLM baseline.

Exemplar Prompting (EP) is a form of visual instruction tuning that augments a pre-trained, instruction-tuned VLM with additional task-specific features for each input. In the original EP, a task-specific model is first trained on a downstream task and its latent representation is treated as a task-specific “exemplar”. During a second-stage instruction tuning, this exemplar feature is projected into the language space and fed to the LLM alongside the original image-derived visual tokens. The VLM's vision encoder is kept frozen, while only the exemplar adaptor and the LLM are fine-tuned, enabling the model to condition its responses jointly on both the generic visual representation and the task-specific exemplar.

In our setting, we adapt this idea by directly using the RexGroundingCT segmentation masks as task-specific features, rather than training a separate task-specific model. After training the baseline MS-VLM on RexGroundingCT for volumetric report generation, we introduce an additional EP stage. For each input CT volume, the corresponding segmentation masks are passed through a lightweight adaptor consisting of a 3D CNN followed by a linear layer. This adaptor produces mask-based features that serve as supplementary visual cues in addition to the original volume-level representation obtained from the frozen MS-VLM vision encoder. The two sets of visual features—volume features and mask-derived exemplar features—are then concatenated and provided to the LLM. During this EP stage, we fine-tune the lightweight segmentation-mask adaptor and the LLM, while keeping the MS-VLM vision encoder fixed. Unlike the original EP, our approach

therefore does not require training a separate task-specific model; the segmentation masks themselves act as direct task-specific exemplars.

As shown in Table 7, MS-VLM (EP) yields consistent improvements over the baseline MS-VLM across both NLG and CA metrics. These results indicate that explicit lesion-aware features derived from segmentation masks help the model align its volumetric representation more closely with the report content. In turn, this reduces the mismatch between predicted abnormalities and their underlying lesions and partially mitigates hallucinations driven by weak or diffuse visual grounding. Overall, this experiment demonstrates that MS-VLM can effectively leverage exemplar prompting with segmentation-derived features to improve report quality.

4.2. Experimental results on rectal MRI

The rectal MRI experiments were conducted to evaluate the proposed MS-VLM architecture under key conditions characterized by fewer slices per volume compared to CT, the necessity of integrating multiple views (e.g., axial, sagittal) and phases (e.g., T1, T2), and limited data availability. These conditions were designed to assess the advantages of MS-VLM's pre-trained 2D vision encoder using DINO and its subsequent Z-former self-supervised learning approach, along with the overall training framework, in efficiently handling multi-phase and multi-view data and adapting to data-scarce clinical settings.

Table 8 summarizes the experimental results, demonstrating that MS-VLM outperformed baseline methods, including CT2Rep and MS-VLM (3D encoder), across all critical findings. Notably, the proposed MS-VLM achieved significant improvements in key metrics such as T stage classification, CRM threatening, and MLNI detection. In these metrics, MS-VLM not only outperformed CT2Rep and MS-VLM (3D encoder) but also exceeded the performance of a variant using only the primary view, T2 axial, within the identical MS-VLM framework. Table 8 also demonstrates the effect of adding individual T1, T2 sagittal volumes to T2 axial have on report generation. Compared to the baseline MS-VLM (T2 axial only), adding T1 sagittal or T2 sagittal individually resulted in trade-offs. Specifically, the prediction accuracy for “location” and “peritoneum involvement” slightly decreased, while other categories showed either no significant change or a mild improvement. However, when both sagittal views (T1 and T2) were used together alongside the T2 axial series, performance improved across nearly all categories, with notable gains outside of T-staging. MS-VLM (all views & phases) which uses extra T1 axial volumes on top of T2 axial + T1/T2 sagittal (combined) then achieves the highest overall performance across most abnormalities, with the exception of a slight decline in “Mesorectal Lymph Node Involvement (MLNI)”. These results suggest that including a single additional view or phase may disturb the model's original spatial representation and potentially hinder performance for specific findings. However, when provided with a sufficiently diverse and comprehensive set of views and phases, MS-VLM can more effectively learn the mapping between volumetric features and the corresponding textual descriptions. Such findings validate MS-VLM's ability to effectively integrate information from multi-view and multi-phase data, underscoring the importance of such integration for optimal performance.

For qualitative examples of the generated reports, refer to Appendix D.

4.3. Results on computational efficiency

Our approach leverages DINO to effectively learn critical visual representations from individual slices in the initial stage, followed by the Z-former to capture inter-slice dependencies. This two-step strategy facilitates rapid convergence during subsequent training with 3D volumetric data, offering computational efficiency in terms of data utilization. Furthermore, our method maximizes the benefits of pre-trained LLMs by separating the tuning process into distinct stages,

Table 8

Comparison of o1-mini evaluation performance and F1 scores of abnormality prediction between MS-VLM and baseline methods.

Model	Location	Peritoneum	T stage	CRM	ASI	MLNI	EMVI
	Accuracy			F1			
CT2Rep (T2 axial)	<u>0.209</u>	<u>0.488</u>	0.163	0.143	<u>0.133</u>	0.108	0.000
MS-VLM (3D encoder, T2 axial)	0.116	0.372	0.116	0.171	0.121	0.426	<u>0.111</u>
MS-VLM (T2 axial)	<u>0.209</u>	0.395	0.279	<u>0.308</u>	0.114	0.298	<u>0.111</u>
MS-VLM (T2 axial + T1 Sagittal)	0.116	0.266	0.333	<u>0.308</u>	0.222	0.426	<u>0.111</u>
MS-VLM (T2 axial + T2 Sagittal)	0.133	0.200	<u>0.467</u>	<u>0.308</u>	0.114	0.426	<u>0.111</u>
MS-VLM (T2 axial + T1/T2 Sagittal)	0.233	<u>0.488</u>	<u>0.467</u>	<u>0.308</u>	0.222	0.762	0.143
MS-VLM (all views & phases)	0.233	0.512	0.581	0.421	0.222	<u>0.667</u>	0.143

Accuracy is evaluated using o1-mini. The best values are highlighted in bold, and the second-best are underlined. The name of the abnormalities are shortened for readability.

Table 9

Comparison of total number of parameters, TFLOPs and inference speed between Merlin and MS-VLM (proposed).

Method	Parameters	TFLOPs	Inference time (s)
Merlin	7048M	0.21	19.8
MS-VLM (proposed)	7919M	0.14	33.0

allowing us to exploit the advantages of pre-training to further improve computational efficiency. To validate this hypothesis, we compared the convergence speed of models during the training process. Fig. 7 demonstrates that the MS-VLM architecture achieves the fastest convergence in CA metric performance compared to other methods. For instance, CT2Rep attains an average F1 score of only 0.038 after the same five epochs, indicating that it requires significantly more training to achieve comparable performance. Specifically, CT2Rep needed 40 epochs of training to reach an average F1 score of 0.175. Notably, the proposed MS-VLM approach, which utilizes a 2D vision encoder and the Z-former, achieves faster convergence and superior performance compared to methods employing a 3D vision encoder. This demonstrates the efficiency of the sequential training strategy, wherein DINO is used to pre-train the 2D vision encoder, followed by training the Z-former to capture inter-slice dependencies. These results suggest the computational advantages of our staged approach, effectively leveraging the strengths of pre-trained LLMs to achieve efficient convergence while maintaining high performance.

We report a comparison of the computational efficiency between MS-VLM and Merlin, as summarized in Table 9. While MS-VLM exhibits lower TFLOPs, it requires a longer inference time than Merlin. This is primarily due to MS-VLM's architecture, which performs iterative forward passes through the DINO vision encoder to generate slice-wise representations before aggregating them into a volumetric feature. As each slice is processed individually, this sequential operation introduces additional latency. Nevertheless, this increase in inference time is unlikely to pose a practical limitation in clinical settings. If we view these models as intelligent assistants designed to aid expert radiologists by generating candidate reports or suggesting potential findings, the added time is minor relative to the entire clinical workflow. In real-world scenarios, the time required to acquire CT scans from patients and transmit them to radiologists typically far exceeds (approximately 5–10 min) the inference time (approximately 30 s) of either model. Furthermore, the inference latency of MS-VLM can be significantly reduced in environments equipped with higher-memory GPUs than the NVIDIA A100 40 GB GPUs we used for this work. (e.g., NVIDIA H100 80 GB). With sufficient VRAM, multiple slices can be processed in parallel by batching along the depth dimension, effectively minimizing the number of iterative forward passes required by the DINO vision encoder.

5. Discussion

Recent advancements in multi-modal AI models and improvements in model scalability have underscored the potential of true multi-modal

medical foundation models (Tu et al., 2024; K. Zhang et al., 2024; Lu et al., 2024). These models aim to integrate diverse data modalities such as images and text, with a growing emphasis on extending capabilities beyond 2D images to volumetric 3D imaging modalities like CT and MRI (Wu et al., 2023; Hamamci et al., 2024a; Chen et al., 2024; F. Bai et al., 2024). Despite these promising developments, the performance of VLMs for 3D medical imaging remains suboptimal, particularly in tasks requiring precise alignment of image features with textual descriptions.

A key limitation lies in the relative under-performance of vision encoders compared to LLMs. While LLMs excel in understanding detailed clinical context and generating coherent reports, vision encoders often struggle to embed fine-grained anatomical details from 3D medical images effectively (Hu et al., 2024). This discrepancy arises from the inherent differences in how current models process 3D volumetric data.

Most existing approaches treat 3D medical images in a manner analogous to videos, employing 3D patch embeddings within 3D vision encoders. However, this methodology overlooks critical distinctions between 3D medical imaging and video data. Videos typically consist of a large number of frames (e.g., a 5-minute video at 30 frames per second results in $224 \times 224 \times 9000$), characterized by substantial redundancy across frames, which makes 3D patch embeddings effective for capturing spatial and temporal correlations (Madan et al., 2024). In contrast, 3D medical images generally contain significantly fewer slices (e.g., $512 \times 512 \times 24$) with lower redundancy between adjacent slices. Under such conditions, the use of 3D patch embeddings can lead to over-correlated representations along the z-axis, potentially impairing model performance. This limitation is particularly pronounced when subtle pathologies appear or disappear between slices, making it crucial to avoid excessive correlation that could obscure meaningful slice-specific information.

To address these challenges, we developed MS-VLM, a VLM architecture inspired by radiologists' workflows. Radiologists typically analyze 3D medical images by examining individual slices sequentially and synthesizing information across slices and views. They also focus on inter-slice relationships; for instance, a structure suspected to be a tumor might be identified as a blood vessel if it continues across slices, whereas a nodular shape that disappears in subsequent slices may be classified as a tumor. Radiologists integrate these observations with multi-view (e.g., axial, sagittal) and multi-phase (e.g., T1, T2) imaging to arrive at a comprehensive interpretation.

MS-VLM mirrors this process by independently embedding slice-specific features using a 2D vision encoder and aggregating them into volumetric representations through the Z-former. This approach significantly reduces redundant correlations along the z-axis while preserving slice-specific granularity and inter-slice relationships, thereby enhancing diagnostic accuracy.

The flexibility of MS-VLM was demonstrated in experiments involving varying CT scan ranges. In scenarios with a larger number of slices, MS-VLM outperformed fixed-length models, which often experience information loss when exceeding their predefined length, by effectively leveraging its ability to aggregate meaningful information across the entire volume without such losses. This adaptability is particularly

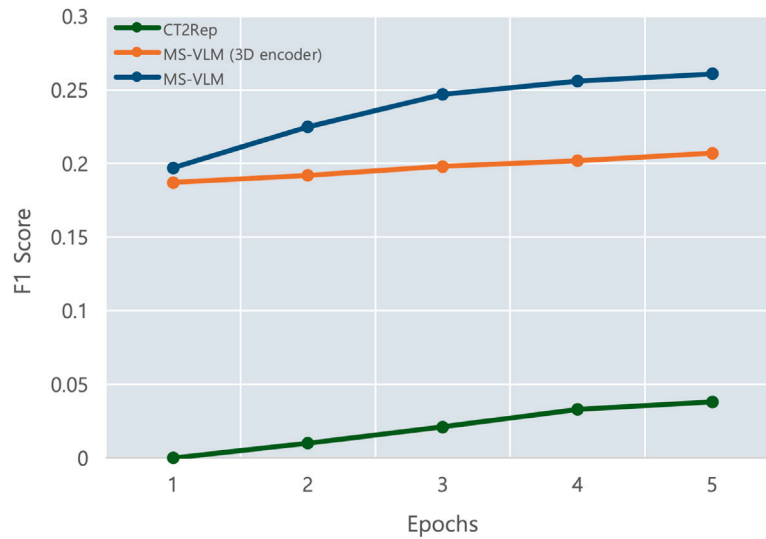


Fig. 7. Average F1 score across 18 chest abnormalities of models over training epochs. MS-VLM achieves superior average F1 score than the baseline methods.

valuable in clinical settings, where scan ranges can vary depending on the target anatomy or the imaging protocol employed.

Moreover, MS-VLM exhibited robust performance in multi-view and multi-phase integration, both of which are critical for comprehensive diagnostic workflows. For instance, in rectal MRI experiments, MS-VLM achieved superior accuracy by simultaneously processing axial and sagittal views alongside multi-phase data, such as T1, T2 images. This capability underscores its clinical relevance, as the integration of multi-view and multi-phase data is essential for identifying critical abnormalities and ensuring precise diagnoses.

In addition, MS-VLM has demonstrated the ability to serve as a valid backbone for downstream segmentation task, by achieving performance comparable to the original SegVol, which uses a 3D ViT pretrained via SimMIM. This capability likely arises from the self-supervised training of the MS-VLM vision encoder, which enables the model to extract more fine-grained and distinctive representations of volumetric data. This is particularly beneficial for downstream vision-centric tasks such as segmentation, bounding box detection, and prediction of abnormality size and location. In contrast, many related works such as 3D-CT-GPT, CT2Rep, and Merlin rely on CLIP-style contrastive learning, which aligns visual representations directly to their paired text embeddings. While this allows for efficient vision-language alignment, it introduces a key limitation: visual features are constrained by the semantic expressiveness of the corresponding report. As a result, visually distinct volumes may be mapped to nearly identical embeddings if the associated reports use similar language. For example, consider two chest CT volumes from different patients, both diagnosed with cardiomegaly, and whose reports simply state: “The CT shows cardiomegaly”. A CLIP-based encoder would likely generate similar embeddings for both, overlooking potentially critical differences in lung size, presence of ground-glass opacities, or other subtle findings. This hinders their utility in downstream visual grounding tasks such as segmentation or localization. MS-VLM, on the other hand, uses DINO to extract slice-level features, which are aggregated by the Z-former to form a volume-level representation. This two-stage process yields more expressive and semantically rich representations that are better suited for diverse downstream applications.

Another significant advantage of MS-VLM is its reliance on 2D vision encoders, which benefit from advanced pre-training techniques like DINO. This design enables the model to converge rapidly and efficiently learn visual representations, demonstrating improved computational efficiency in terms of fast convergence, as shown in our experiments. Furthermore, MS-VLM’s architecture is highly generalizable, capable of adapting to various imaging modalities, including CT, MRI, PET-CT, and 3D ultrasound. This adaptability positions MS-VLM as a versatile

tool for diverse clinical applications, ranging from diagnostic imaging to complex multi-modal analyses.

Despite its promising results, MS-VLM has several limitations. First, the model is not entirely free from hallucination errors and may occasionally miss critical findings. [Appendix F](#) demonstrates the two major sources of hallucination induced by MS-VLM for chest CT report generation: Overfitting to frequently occurring abnormalities and Hallucination propagation driven by semantic correlation with preceding false-positive findings. A large portion of hallucinations take the form of false-positive findings, most commonly involving “lung nodules”—the most abundant abnormality class in CT-RATE. Since LLMs are trained with a next-token prediction objective, they tend to memorize and reproduce frequently occurring sequences in the training data. This creates a bias during instruction fine-tuning, leading the model to associate visual inputs disproportionately with common abnormalities. As a result, the LLM may overfit to dominant classes, increasing the risk of generating false-positive findings. For example, as shown in [Appendix F](#), MS-VLM incorrectly reports “millimetric non-specific nodules in both lungs” when the correct diagnosis is emphysematous changes. This type of hallucination highlights the impact of imbalanced abnormality distribution in training data. The second type of hallucination involves semantic propagation of earlier false positives. In the second example from [Appendix F](#), the LLM first hallucinates an aortic abnormality and subsequently adds false-positive findings related to the atrium, despite no atrial abnormalities being mentioned in the ground truth or present in the training dataset. This behavior likely stems from the LLM’s learned tendency to generate medically correlated terms in sequence, drawing on its prior knowledge rather than actual visual grounding. The model appears to complete familiar clinical patterns rather than interpreting the visual input independently for each finding. While this study introduces a novel proof-of-concept architecture, it is important to note that larger-scale studies utilizing more diverse and higher-quality datasets could potentially achieve better absolute performance and mitigate aforementioned false-positive findings.

Second, the current evaluation of MS-VLM is limited both in terms of modality coverage and the granularity of visual grounding. In this study, we validated the model only on chest CT and rectal MRI. As a result, its generalizability to other imaging modalities (e.g., abdominal CT) remains unexplored. We plan to further develop MS-VLM by leveraging recent large-scale benchmarks such as MedTrinity-25M ([Xie et al., 2024](#)) to assess its applicability across a broader range of clinical tasks and modalities, and to obtain richer supervision for more fine-grained visual grounding.

As illustrated in [Table 6](#) and [Appendix G](#), the current explainability of MS-VLM relies on analyzing the attention distribution between the

learnable queries in the perceiver resampler and the set of slice-level visual representations. These results suggest that MS-VLM does not rely solely on global volume-level statistics. Instead, it learns abnormality-specific attention over slices that reflects key clinical characteristics: higher slice coverage for diffuse abnormalities (e.g., emphysema), more localized yet still meaningful attention for focal lesions (e.g., nodules), and performance reductions in settings where even human experts face substantial ambiguity (e.g., subtle airway or septal thickening). We believe this behavior supports the interpretability of MS-VLM's attention mechanism and demonstrates that its slice-level attention patterns are aligned with radiological intuition.

However, this attention-based analysis is limited to identifying which slices are prioritized and does not provide fine-grained localization of abnormalities within individual slices. As future work, we plan to incorporate slice-level region-of-interest (ROI) annotations from large-scale benchmarks such as MedTrinity-25M to enable MS-VLM to learn explicit spatial mappings between abnormalities and their intra-slice locations, thereby providing a more detailed interpretability analysis at the voxel or lesion level. Complementing this direction, Table 7 shows that adding abnormality segmentation-mask-derived features via exemplar prompting improves both NLG and CA metrics by enhancing volume-to-report alignment. This empirical gain suggests that stronger lesion-level visual grounding not only improves interpretability, but also leads to better vision-to-text alignment and overall report quality, reinforcing the importance of explicit lesion-aware supervision in future extensions of MS-VLM.

Third, this study did not include direct expert evaluation, which would provide critical insights into the model's clinical utility and guide its refinement for practical deployment.

6. Conclusion

In this paper, we introduced a MS-VLM by introducing a novel Z-former architecture that integrates slice-by-slice embeddings into a volume representation for volumetric 3D medical image interpretation. MS-VLM effectively addresses the challenges of processing 3D medical imaging by adopting a radiologist-inspired approach that preserves slice-specific granularity and captures inter-slice relationships. Its ability to integrate multi-view and multi-phase information, combined with flexibility across varying scan ranges and transferability to vision-centric task, demonstrates its adaptability to diverse clinical scenarios. These advancements position MS-VLM as a significant step forward in advancing medical VLMs, particularly by enhancing vision encoders to process the complexity of 3D medical imaging with broad generalizability across various modalities.

CRediT authorship contribution statement

Changsun Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Sangjoon Park:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Cheong-Il Shin:** Writing – review & editing, Validation, Supervision, Formal analysis. **Woo Hee Choi:** Writing – review & editing, Validation, Supervision, Formal analysis. **Hyun Jeong Park:** Writing – review & editing, Validation, Supervision, Formal analysis. **Jeong Eun Lee:** Writing – review & editing, Validation, Supervision, Resources, Data curation. **Jong Chul Ye:** Writing – review & editing, Supervision, Resources, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Details on synthetic VQA generation

Inspired by LLM-CXR (Lee et al., 2023), we generate an instruction prompt that guides LLaMA-3-8B to generate synthetic VQA pairs from a radiology report (see Fig. A.8). Fig. A.9 shows an example from RadGenome-ChestCT dataset used for MS-VLM training.

Appendix B. Availability of rectal MRI dataset

The in-house rectal MRI dataset consists of volumes acquired using three different contrasts: T1, T2 and DWI. For T1, T2 volumes, each volume was taken with one of Turbo Spin Echo (TSE), Fast Spin Echo (FSE) and Fast recovery FSE (FrFSE) imaging techniques. Only the T1, T2 volumes were used for training MS-VLM (see Table B.10).

Appendix C. Instruction prompts for LLM-based evaluation

Fig. C.10 shows the instruction prompt for GPT-4o-mini used to evaluate the generated reports on the CT-RATE test set. Fig. C.11 shows the instruction prompt for o1-mini used to evaluate the generated reports on the rectal MRI test set.

Appendix D. Qualitative examples

Figs. D.12 and D.13 show examples of the qualitative evaluation on CT-RATE. Fig. D.14 shows an example of VQA performed by MS-VLM on CT-RATE. Figs. D.15 and D.16 show examples of the qualitative evaluation on rectal MRI. Figs. D.12, D.13, D.15 and D.16 demonstrate the superiority of combining a 2D vision encoder with the Z-former over using a single conventional 3D vision encoder.

Appendix E. CA metrics on chest CT report generation & VQA

Tables E.11 and E.12 show the CA metrics across 18 chest-related abnormalities for generated reports and VQA on the CT-RATE test set, respectively.

Appendix F. Representative examples of hallucination in MS-VLM

Fig. F.17 shows representative examples of two major sources of hallucination in MS-VLM for chest CT report generation: false-positive findings associated with frequent abnormalities and false-positive findings induced by semantic correlation.

You are a radiologist asking questions about a 3D chest CT scan. You always give long, detailed explanations as answers.

You are given a 3D chest CT scan, delimited by triple backticks.

Create questions about the given 3D chest CT scan, each question followed by an appropriate answer.

Each question should ask whether the scan is positive for any of the abnormalities given in the following list.

['Medical material', 'Arterial wall calcification', 'Cardiomegaly', 'Pericardial effusion', 'Coronary artery wall calcification', 'Hiatal hernia', 'Lymphadenopathy', 'Emphysema', 'Atelectasis', 'Lung nodule', 'Lung opacity', 'Pulmonary fibrotic sequela', 'Pleural effusion', 'Mosaic attenuation pattern', 'Peribronchial thickening', 'Consolidation', 'Bronchiectasis', 'Interlobular septal thickening']

Output your questions and answers in JSON format like:

```
[{"question": "<question1>", "answer": "<answer1>"},
{"question": "<question2>", "answer": "<answer2>"}, ...].
```

...

```
{report}
```

...

Note, questions are never about any change from the last or previous chest CT scans. Questions are also never about future plans; questions always focus on the abnormalities listed above. Answers are very detailed and include explanations without repeating words that are in the question.

Here are the pairs of questions and answers:

Fig. A.8. Instruction prompt given to LLaMA-3-8B for synthetic VQA generation from CT-RATE reports.

Q: What abnormalities are present in the aorta?
A: Atheromatous plaques are present in the aorta.

Fig. A.9. An example of a QA pair from RadGenome-ChestCT dataset, which asks about the type of abnormalities present in a certain region.

Table B.10

Availability of in-house rectal MRI dataset.

Contrast	Imaging technique	Modifier	Imaging plane		
			Axial (+oblique)	Sagittal	Coronal (+oblique)
T1-weighted	TSE/FSE	None	✓	✓	✗
	TSE/FSE	FS/SPIR, CE	✓	✓	✓
T2-weighted	TSE/FSE/FrFSE	None	✓	✓	✓
DWI	ADC	N/A	✓	✗	✗
	b-value 0	N/A	✓	✗	✗
	b-value (800 or 1000)	N/A	✓	✗	✗

Appendix G. Visualization of learnable query attention to slice representations

Fig. G.18 shows a heatmap of learnable query attention over slice representations for a chest CT volume containing abnormalities, whereas Fig. G.19 shows the corresponding heatmap for a chest CT volume

without abnormalities. Figs. G.18 and G.19 illustrate that MS-VLM assigns strong attention to slice representations associated with abnormal slices, while distributing attention more uniformly across slices in the absence of abnormalities. This observation indicates that MS-VLM learns to focus on lesion-containing slices for accurate abnormality prediction.

The following is the ground truth radiology report for a chest CT scan: {gt_report} The following is the radiology report generated by the model: {ours_report} Compare the model-generated report with the ground truth, and evaluate if the mention of abnormalities is accurate in terms of their presence or absence. Evaluation Criteria: 0 = Incorrect 1 = Correct Output the answer as a single value 0 or 1.
The following is the ground truth radiology report for a chest CT scan: {gt_report} The following is the radiology report generated by the model: {ours_report} Compare the model-generated report with the ground truth, and evaluate if the mention of the location of abnormalities is accurate. Evaluation Criteria: 0 = Incorrect 1 = Correct Output the answer as a single value 0 or 1.
The following is the ground truth radiology report for a chest CT scan: {gt_report} The following is the radiology report generated by the model: {ours_report} Compare the model-generated report with the ground truth, and evaluate if the mention of the severity of abnormalities is accurate. Evaluation Criteria: 0 = Incorrect 1 = Correct Output the answer as a single value 0 or 1.
The following is the ground truth radiology report for a chest CT scan: {gt_report} The following is the radiology report generated by the model: {ours_report} Compare the model-generated report with the ground truth, and evaluate if there are any hallucinations (false findings) present that were not mentioned in the ground truth report. Evaluation Criteria: 0 = Incorrect (Hallucination present) 1 = Correct (No hallucination) Only output the evaluation score 0 or 1 as the answer.

Fig. C.10. Instruction prompt for GPT-4o-mini used to evaluate the generated reports on the CT-RATE test set.

Table E.11

Precision, Recall, and F1 score across 18 chest-related abnormalities on the generated reports by MS-VLM.

Abnormality	Precision	Recall	F1 score
Medical material	0.155	0.158	0.157
Arterial wall calcification	0.306	0.367	0.334
Cardiomegaly	0.249	0.298	0.271
Pericardial effusion	0.130	0.214	0.162
Coronary artery wall calcification	0.254	0.435	0.321
Hiatal hernia	0.141	0.245	0.179
Lymphadenopathy	0.268	0.393	0.319
Emphysema	0.215	0.410	0.282
Atelectasis	0.257	0.305	0.279
Lung nodule	0.466	0.536	0.499
Lung opacity	0.440	0.592	0.505
Pulmonary fibrotic sequela	0.256	0.451	0.327
Pleural effusion	0.133	0.315	0.187
Mosaic attenuation pattern	0.128	0.230	0.164
Peribronchial thickening	0.133	0.353	0.194
Consolidation	0.235	0.370	0.287
Bronchiectasis	0.125	0.100	0.111
Interlobular septal thickening	0.104	0.157	0.125
Average	0.222	0.329	0.261

You are a radiologist who needs to evaluate the correctness of VLM report (vlm_report) compared to the Ground Truth (GT) report. You have seven categories to evaluate in the following order.

1) Evaluate whether the VLM report accurately specifies the 'tumor location', particularly the 'lowest tumor margin from the anal verge,' as described in the Ground Truth (GT) report. The reported distance should closely match or explicitly reference a location similar to the GT. Refer to the example below.

Example:
 Correct: GT: "The lowest tumor margin is situated approximately 4.7 cm from the anal verge." VLM: "The lowest tumor margin is about 4.3 cm from the anal verge."
 Incorrect: GT: "The lowest tumor margin is approximately 6.4 cm from the anal verge." VLM: "The lowest tumor margin is approximately 2.7 cm from the anal verge."

2) Assess whether the 'tumor relationship with the peritoneum' in the VLM report is accurately described as in the Ground Truth (GT) report. Terms such as 'no relationship' or 'partially peritonealized' should match precisely. Any omission of peritoneal involvement details will be marked as negative. See the example below.

Example:
 Correct: GT: "The tumor is partially peritonealized in its relationship to the peritoneum." VLM: "The tumor has partial peritonealization."
 Incorrect: GT: "There is no relationship between the tumor and the peritoneum." VLM: "The tumor is partially peritonealized in its relationship to the peritoneum."

3) Verify if the VLM report accurately specifies the 'Clinical T stage' as defined in the Ground Truth (GT) report. Precise T-stage notation, such as 'CT2' or 'CT3,' is required. Refer to the example below.

Example:
 Correct: GT: "The tumor is classified as CT2 in T-staging." VLM: "T-stage classification: CT2."
 Incorrect: GT: "Classified as CT3." VLM: "The tumor is classified as CT2 in T-staging."

4) Confirm whether the 'Circumferential resection margin (CRM) involvement' is specified in the VLM report as it is in the Ground Truth (GT) report. CRM involvement should be clearly stated as positive or negative. If not mentioned, CRM involvement is considered to be negative. Refer to the example below.

Example:
 Correct: GT: "Negative for circumferential resection margin involvement." VLM: "No CRM involvement detected."
 Incorrect: GT: "Positive CRM involvement." VLM: "No CRM involvement detected."

Fig. C.11. Instruction prompt for o1-mini used to evaluate the generated reports on the rectal MRI test set. The generated report is evaluated on seven abnormalities: Tumor location, Peritoneal involvement, T staging, CRM, ASI, MLNI, and EMVI.

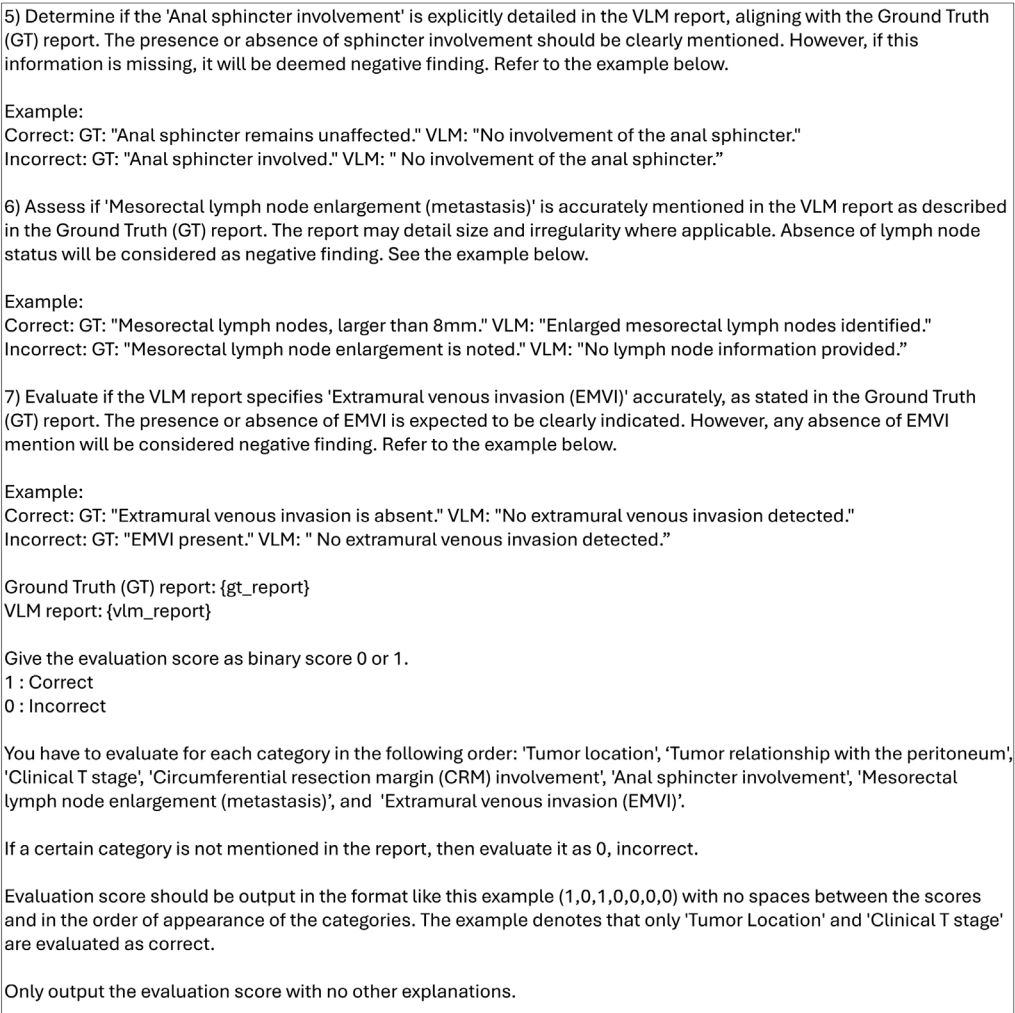


Fig. C.11. (continued).



True Positive / True Negative / False Positive / False Negative

Ground Truth: CTO is within the normal range. Calibration of the main mediastinal vascular structures is natural. No lymph node with pathological size and configuration was detected in the mediastinum. No pathological size and configuration lymph nodes were detected at both hilar levels. Thoracic esophagus calibration was normal and no significant tumoral wall thickening was detected. Mild hiatal hernia is observed. When examined in the lung parenchyma window; Both hemithorax are symmetrical. The calibration of the trachea and main bronchi is normal and their lumens are clear. Mild thickening of the peribronchial sheath is observed. There are sequelae changes at the apical level. On the right, a nonspecific nodule with a diameter of 2 mm is observed superposed on the minor fissure. Sequelae changes are observed in the middle lobe. There are sequelae changes at the anterobasal level. A subpleural 2mm diameter nodule is observed in the upper lobe posterior segment lateral in the right lung. Pleuroparenchymal sequelae changes are observed in the inferior lingular segment. There is a 2 mm diameter nodule in the apicoposterior segment of the left lung upper lobe.

CT2Rep: Trachea both main bronchi are open. Mediastinal main vascular structures heart contour size are normal. thoracic aorta diameter is normal. Pericardial effusion-thickening was not observed. thoracic esophagus calibration was normal and no significant tumoral wall thickening was detected. No enlarged lymph nodes in prevascular pre-paratracheal subcarinal or bilateral hilar-axillary pathological dimensions were detected. When examined in the lung parenchyma window in both lungs patchy ground glass densities are observed mostly in the lower lobes. The findings were initially evaluated in favor of covid-19 viral pneumonia.

MS-VLM (3D encoder): Trachea and main bronchi are open. Right upper-lower paratracheal millimetric lymph node is observed. No pathological LAP was detected in the mediastinum. The heart and mediastinal vascular structures have a natural appearance. Pleural effusion-thickening was not detected in both hemithorax. In the evaluation of both lung parenchyma; A subpleural nodule with a diameter of 3 mm is observed in the middle lobe of the right lung, which does not differ from the previous examination...

MS-VLM: CTO is within normal limits. Calibration of major vascular structures in the mediastinum was observed to be natural. Thoracic esophageal calibration was followed naturally. No lymph node with pathological size and configuration was detected at the hilar level or in both axillary regions as far as can be seen in non-contrast examination. When examined in the lung parenchyma window; A subpleural 3 mm diameter nodule is observed in the left lung adjacent to the fissure laterally. There are sequelae changes in bilateral upper lobe apical segments. Sequelae changes are also present in bilateral lower lingula and inferior lingular segments. Density compatible with a millimetric calcific pulmonary nodule is observed...

Fig. D.12. Qualitative evaluation on CT-RATE. Each sentence from the report is color coded to identify True Positive, True Negative, False Positive and False Negatives on the findings mentioned in the ground truth report. MS-VLM achieves greater recall on the true positives.



True Positive / True Negative / False Positive / False Negative

Ground Truth: Mediastinal structures were evaluated as suboptimal since the examination was unenhanced. As far as can be seen trachea and lumen of both main bronchi are open. No occlusive pathology was detected in the trachea and lumen of both main bronchi. An image of a catheter extending superiorly to the vena cava was observed. Heart contour size is natural. Pericardial thickening-effusion was not detected. Calcified atherosclerotic changes were observed in the wall of the thoracic aorta... A few calcified lymph nodes with a short axis smaller than 1 cm were observed in the left hilar region. In addition lymph nodes measuring 1 cm in the short axis of the largest were observed... When examined in the lung parenchyma window interlobular septal thickenings and alveolar consolidation areas were observed in the upper lobe of the left lung. There are patches of ground glass density increases in both lungs. a few parenchymal nodules the largest of which was 8 mm in diameter were observed in the right lung...

CT2Rep: Trachea both main bronchi are open. Mediastinal main vascular structures heart contour size are normal. Thoracic aorta diameter is normal. Pericardial effusion-thickening was not observed... When examined in the lung parenchyma window widespread patchy ground glass densities and consolidation areas are observed in both lungs. the findings were initially evaluated in favor of covid-19 viral pneumonia.

MS-VLM (3D encoder): Trachea and both main bronchi are open. No occlusive pathology was detected in the trachea and both main bronchi. There is minimal peribronchial thickening in both lungs, more prominent on the right. In addition, nodular ground glass areas-consolidations accompanied by interlobular septal thickenings were observed in a small area in both lung lower lobes, most prominently in the posterobasal segment of the left lung lower lobe... Both lungs have millimetric calcific nodules. Mediastinal structures cannot be evaluated optimally because contrast material is not given. As far as can be seen; Heart contour and size are normal. The widths of the mediastinal main vascular structures are normal. Minimal pleural effusion was observed on the left. Pericardial effusion was not detected. Pleural paraseptal emphysematous changes are present in both lungs.

MS-VLM: Mediastinal structures were evaluated as suboptimal since the examination was unenhanced. As far as can be seen trachea both main bronchi are open. No obstructive pathology was detected in the lumen of the trachea and lobar and segmental lumens of both main bronchi. Heart contour size is normal. Pericardial thickening-effusion was not observed... Calcified atherosclerotic changes are observed on the wall of the coronary artery. Multiple reactive millimetric lymph nodes with a short axis smaller than 1 cm were observed in the left inferior pulmonary prevascular area. When examined in the lung parenchyma window peripherally located crazy paving pattern accompanied by patchy consolidation areas containing air bronchograms and ground glass densities were observed in both lungs.

Fig. D.13. Qualitative evaluation on CT-RATE. Each sentence from the report is color coded to identify True Positive, True Negative, False Positive and False Negatives on the findings mentioned in the ground truth report. MS-VLM achieves greater recall on the true positives while generating less false positive findings.

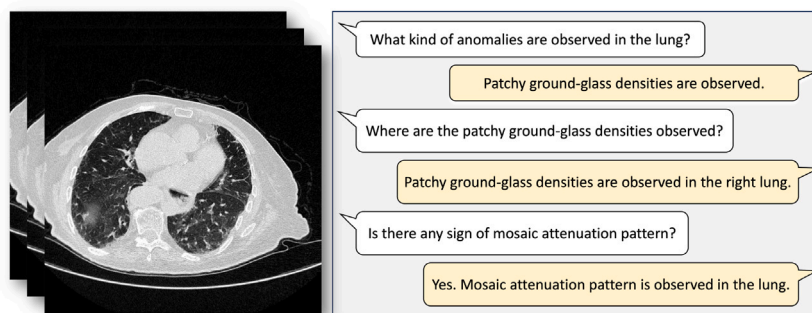


Fig. D.14. An example of VQA by MS-VLM on a test set sample of CT-RATE.

Ground Truth: The lowest tumor margin is about 3.6 cm from the anal verge. There is no relationship between the tumor and the peritoneum. The tumor is partially encircling in its circumferential location and measures about 4.5 cm longitudinally. It is classified as CT2 in T-staging. There is no involvement of the circumferential resection margin or the anal sphincter. Mesorectal lymph nodes are absent, and there is no evidence of extramesorectal lymph nodes, nor any evidence of extramural venous invasion (EMVI). MS-VLM (3D encoder): The distance of the lowest tumor margin from the anal verge is about 7.5 cm, with no relationship to the peritoneum. It is partially encircling in its circumferential location and measures about 1.2 cm longitudinally. In T-staging, it is classified as CT3 because it exceeds 5 mm in thickness. There is involvement by positive circumferential resection margin status (at 2:45 hours) but without positive extramesorectal fascia. Anal sphincter involvement includes external sphincter at the levator ani muscle; internal sphincter involves both the puborectalis and ciliaris externus; mesorectal lymph nodes are absent while there are inguinal lymph nodes present. Additionally, there is a benign biologic lesion described as an enlarged obturator internus. MS-VLM: The distance of the lowest tumor margin from the anal verge is about 8.2 cm, and there is no relationship between the tumor and the peritoneum. The circumferential location of the tumor is partially encircling, with a longitudinal size of approximately 3.4 cm. T-staging indicates a T1b classification, as the thickness of the tumor wall is less than 5 mm. There is no involvement of the circumferential resection margin or the anal sphincter. Mesorectal lymph nodes are present but are smaller than 8mm. There is no evidence of extramesorectal lymph node involvement, while extramural venous invasion (EMVI) is absent.
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Fig. D.15. Qualitative evaluation on rectal MRI. Correct predictions are highlighted. MS-VLM generates reports more faithful to the ground truth in terms of abnormality prediction.

Ground Truth: The distance between the lowest tumor margin and the anal verge is about 7.6 cm. The tumor partially involves the peritoneum. It is completely encircling the lumen. It measures about 4.5 cm longitudinally. T-staging indicates it as CT4a, greater than CT3 (> 5mm), and there is involvement of the circumferential resection margin. The anal sphincter is absent. There are mesorectal lymph nodes present and the borders are irregular. Extramural venous invasion (EMVI) is absent. MS-VLM (3D encoder): The lowest tumor margin is about 7 cm from the anal verge. There are no relationships between the tumors and the peritoneum. T-staging indicates t1/2, meaning that there is a greater than 50% reduction in size for any involvement by T-staging. Circumferential resection margin involvement shows positive ct3 (>5mm) with negative circ-3r status. Mesorectal lymph node involvement presents as nodes larger than 8 mm with irregular borders. MS-VLM: The lowest tumor margin is situated approximately 7.5 cm from the anal verge, and there is no relationship between the tumor and the peritoneum. It is classified as a T4b tumor because it has invaded the rectal wall completely. In T-staging, circumferential resection margin involvement is noted at the 2 o'clock orientation, indicating positive margins. There are present mesorectal lymph nodes (> 8 mm with irregular borders) but no extramesorectal lymph nodes or extramural venous invasion (EMVI).

Fig. D.16. Qualitative evaluation on rectal MRI. Correct predictions are highlighted. MS-VLM generates reports more faithful to the ground truth in terms of abnormality prediction.

Table E.12

Precision, Recall, and F1 score across 18 chest-related abnormalities on VQA by MS-VLM.

Abnormality	Precision	Recall	F1 score
Medical material	0.132	0.158	0.144
Arterial wall calcification	0.546	0.436	0.485
Cardiomegaly	0.143	0.198	0.166
Pericardial effusion	0.674	0.602	0.636
Coronary artery wall calcification	0.250	0.435	0.317
Hiatal hernia	0.226	0.289	0.254
Lymphadenopathy	0.293	0.537	0.379
Emphysema	0.232	0.346	0.278
Atelectasis	0.279	0.379	0.321
Lung nodule	0.466	0.536	0.499
Lung opacity	0.440	0.592	0.505
Pulmonary fibrotic sequela	0.265	0.368	0.308
Pleural effusion	0.200	0.287	0.236
Mosaic attenuation pattern	0.105	0.185	0.134
Peribronchial thickening	0.133	0.180	0.153
Consolidation	0.252	0.292	0.271
Bronchiectasis	0.178	0.096	0.125
Interlobular septal thickening	0.132	0.156	0.143
Average	0.277	0.337	0.297

1) False-positive finding on frequent abnormality

Candidate: trachea both main bronchi are open. mediastinal main vascular structures heart contour and size are normal. thoracic aorta diameter is normal. pericardial effusion-thickening was not observed. no enlarged lymph nodes in prevascular pre-paratracheal subcarinal or bilateral hilar-axillary pathological dimensions were detected. **when examined in the lung parenchyma window no active infiltration or mass lesion was detected in both lungs. there are millimetric non-specific nodules in both lungs.** no pleural effusion was observed. upper abdominal organs included in the sections are normal...

Ground Truth: The mediastinal main vascular structures and the heart could not be evaluated optimally due to the lack of IV contrast, and as far as can be observed, the calibration of the vascular structures, the heart contour and size are natural. No pericardial, pleural effusion or thickening was detected. Trachea, both main bronchi are open and no occlusive pathology is detected. No pathological increase in thoracic esophagus wall thickness is observed. In the mediastinum, in both axillary regions and in the supraclavicular fossa, no lymph nodes are observed in pathological size and appearance. **When examined in the lung parenchyma window; There are paraseptal-centracinar emphysematous changes in both lungs...**No lytic-destructive lesion was observed in the bone structures within the image. There are degenerative changes. and vertebral corpus heights are preserved.

2) False-positive finding via semantic correlation

Candidate: trachea and both main bronchi were in the midline and no obstructive pathology was observed in the lumen. the mediastinum could not be evaluated in the non-contrast examination. **as far as can be seen the anterior-posterior diameter of the ascending aorta was 4.2 cm and the anterior-posterior diameter of the descending aorta was 3.3 cm.** the diameters of the pulmonary arteries increased. heart contour and size are normal. pericardial and pleural effusion was not observed. **the left atrium is larger than the right atrium and the left ventricle is larger than the right ventricle.** thoracic esophagus calibration was normal. no significant pathological wall thickening was detected. no enlarged lymph nodes in the mediastinum or bilateral hilar regions were detected...

Ground Truth: Trachea, both main bronchi are open. Mediastinal main vascular structures, heart contour, size are normal. Thoracic aorta diameter is normal. Pericardial effusion-thickening was not observed. No enlarged lymph nodes in prevascular, pre-paratracheal, subcarinal or bilateral hilar-axillary pathological dimensions were detected...

Fig. F.17. Representative examples of hallucination in MS-VLM for chest CT report generation.

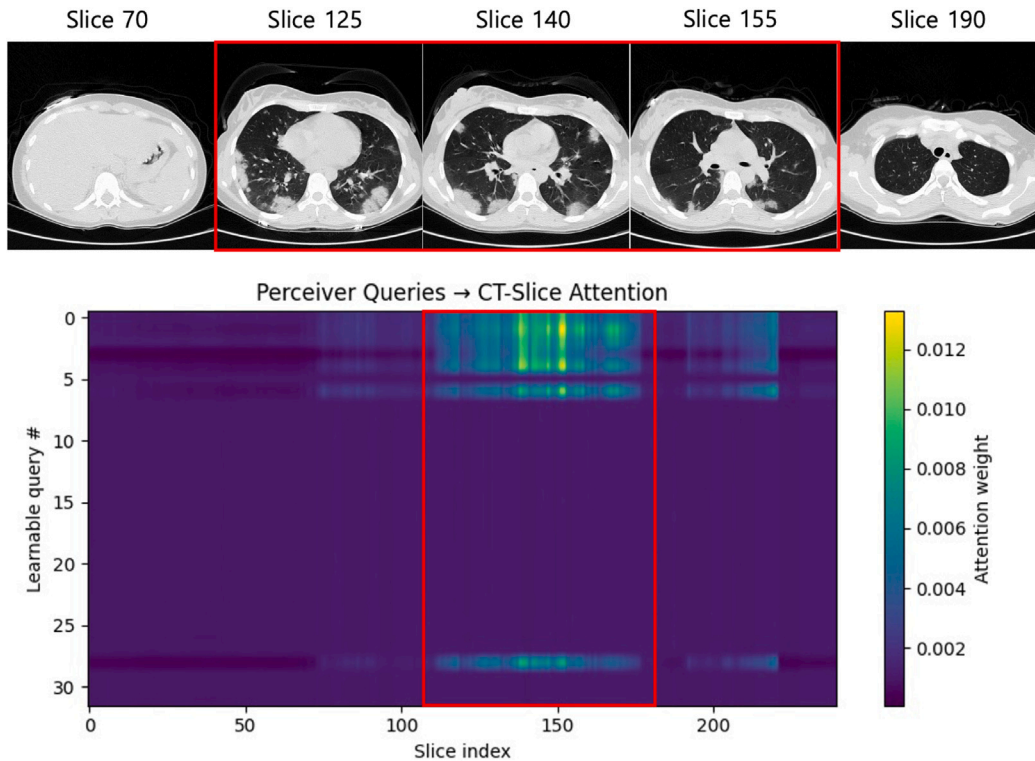


Fig. G.18. Heatmap of learnable query attention over slice representations in the final hidden layer of the perceiver resampler. Slices within the red box contain abnormalities and receive the highest attention, indicating the model's ability to localize relevant regions.

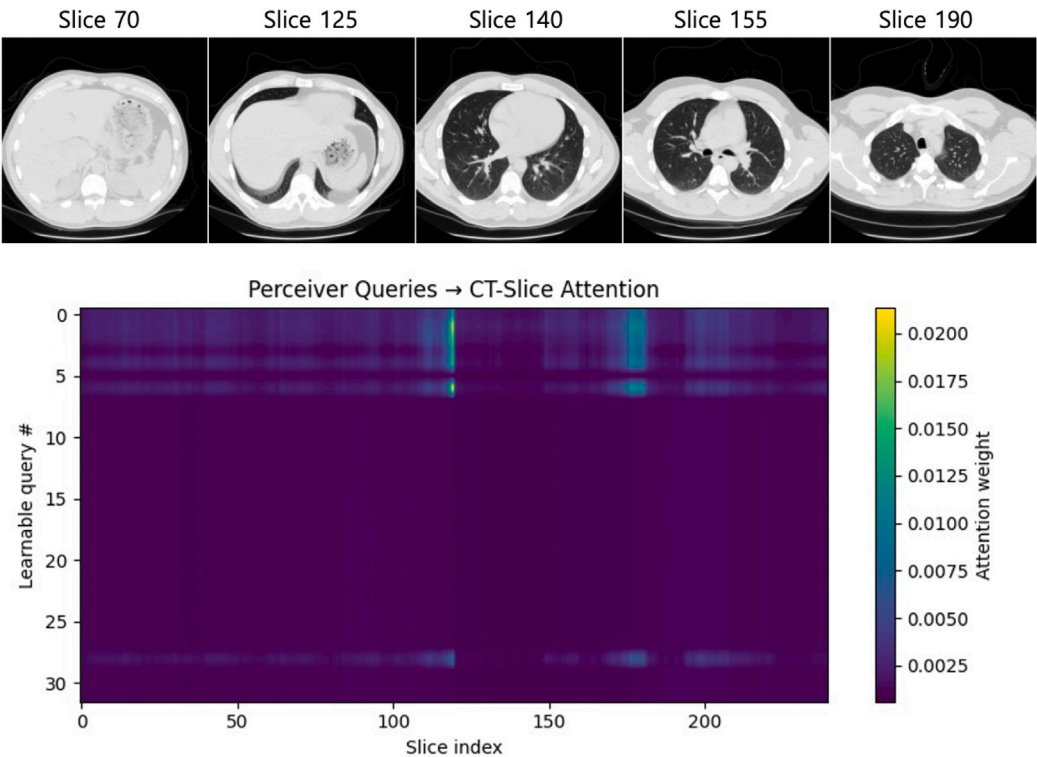


Fig. G.19. Heatmap of learnable query attention over slice representations in the final hidden layer of the perceiver resampler. When MS-VLM receives a volume with no abnormalities, the attention is uniformly distributed across slices, indicating the absence of specific regions requiring focus.

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